

Semantic burstiness across principal directions of textual variation

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Abstract

Long-range correlations in written texts extend over scales far beyond local syntax, but their relation to semantic organisation remains incompletely understood. Previous work has linked these correlations to the bursty recurrence of topically relevant words. Here we generalise that construction from individual lexical items to distributed semantic observables. We map a text token by token onto a trajectory of pre-trained word embeddings, use principal component analysis to identify text-specific directions of semantic variation, and threshold the projected series to define inter-event times; their coefficients of variation form a component-resolved burstiness spectrum $\{B_k\}$. Across 16 literary and scientific texts and the first 20 components, the originals remain systematically above the geometric inter-event-time baseline and above two complementary controls: a word-shuffled null, which preserves the embedding covariance, the PCA eigensystem and the marginal projected distributions while destroying temporal order, and a fractional-Gaussian-noise surrogate, which preserves the Zipfian frequency structure and matches the DFA exponent of the original rank sequence. At $q = 0.95$ the original exceeds both controls in every text-component pair, while the two controls are statistically indistinguishable from each other. Autocorrelation and detrended fluctuation analysis of the semantic projections separate the originals from both controls in the same way. Global rank-level long-range correlations are therefore insufficient to generate the bursty recurrence observed along semantic directions, which instead requires ordered topical organisation.

Keywords: long-range correlations; burstiness; word embeddings; PCA; surrogate data; semantic trajectories; complex systems; topical structure

1 Introduction

1.1 Long-range correlations in texts

Written texts exhibit statistical structure over scales far beyond local syntactic dependencies. Among the most robust observations is the presence of long-range correlations in symbolic sequences derived from natural language. When a text is encoded as a numerical time series

and analysed with detrended fluctuation analysis (DFA) (Peng et al., 1994), the fluctuation function commonly scales as $F(L) \sim L^\alpha$, with $\alpha \in [0.6, 0.8]$, above the uncorrelated value $\alpha = 0.5$, across languages, genres, and encoding choices (Ebeling and Neiman, 1995; Montemurro and Pury, 2002).

This observation raises a fundamental question: what aspect of textual organisation produces memory over hundreds or thousands of tokens? At least two statistical structures must be distinguished. The first is static lexical composition, represented by the vocabulary and word-frequency distribution. Its most familiar regularity is Zipf’s law (Zipf, 1949), according to which the frequency $f(r)$ of the word of rank r follows $f(r) \sim r^{-\gamma}$ with $\gamma \simeq 1$. The second is the temporal ordering of words, through which syntax, discourse, argument, and thematic development unfold. A word-shuffled text preserves lexical composition and Zipf’s law but destroys temporal organisation, causing the DFA exponent to approach $\alpha = 0.5$ (Ebeling and Neiman, 1995; Montemurro and Pury, 2002).

The global DFA exponent, however, does not by itself identify the semantic mechanism responsible for long-range correlations. The surrogate introduced in Degli Esposti and Montemurro (2025) maps fractional Gaussian noise (FGN) onto the empirical rank-frequency structure and is tuned to reproduce the DFA exponent of the original rank sequence. It therefore provides a stationary, rank-level route to the same global scaling and allows us to ask whether matching long-range memory in frequency-rank space is sufficient to reproduce temporal organisation in semantic space.

A geometric approach to long-range textual structure was introduced by Alvarez-Lacalle et al. (2006). They constructed a co-occurrence representation within a sliding “window of attention” and used singular value decomposition to identify text-specific concept vectors. The motion of the sliding window through concept space produced trajectories with power-law autocorrelation over long textual scales, and surrogate tests connected this behaviour to hierarchical textual organisation. Their work established that semantic dynamics can be studied as motion through a reduced concept space.

Our approach shares this geometric viewpoint but differs in three respects. First, we use pre-trained static word embeddings rather than a co-occurrence matrix estimated from a single text. Second, the trajectory is resolved token by token rather than through an aggregated sliding window. Third, our primary observable is the burstiness of threshold crossings along semantic directions, measured through the inter-event-time coefficient of variation, together with the autocorrelation and DFA of the projected series.

1.2 The Altmann et al. mechanism

A mechanism connecting topical organisation to long-range correlations was developed by Altmann et al. (2012). Their framework treats language as a hierarchy of observables, from topics to words and then to lower-level symbolic representations. Correlations can propagate down this hierarchy, whereas burstiness is visible most clearly at semantically relevant levels and can be lost under projection to lower levels.

Formally, a local matching condition α maps a text to a binary sequence $u_t^{(\alpha)} \in \{0, 1\}$. For example, $u_t^{(\alpha)} = 1$ may indicate that token t is the word “prince”. The gaps τ between successive events define an inter-event-time (IET) distribution, whose coefficient of variation

$$\text{CV} = \frac{\sigma_\tau}{\langle \tau \rangle} \quad (1)$$

measures temporal clustering. Topically relevant words tend to occur in concentrated textual domains and therefore show broad IET distributions and elevated burstiness,

whereas common function words are distributed more regularly even when long-range correlations remain visible in lower-level observables.

This framework suggests a natural extension. A topic such as war, philosophical reflection, biological variation, or social interaction is not expressed by one keyword alone. It activates a distributed set of semantically related words. A suitable topic-level observable should therefore respond to a region or direction of semantic space rather than to the exact recurrence of a single word type.

1.3 Motivation and contribution

We represent a text as an embedding trajectory $(\mathbf{e}_t)_{t=1}^M$ in $\mathbb{R}^d$, where each retained token is mapped to a pre-trained static word vector (Mikolov et al., 2013; Pennington et al., 2014). PCA of this trajectory identifies an ordered set of orthogonal, text-specific directions of semantic variation. Projecting the trajectory onto a direction $\mathbf{v}_k$ yields a continuous semantic time series; thresholding that series produces a binary event sequence to which the IET construction of Altmann et al. (2012) can be applied. The resulting spectrum $\{B_k\}_{k=1}^K$ measures burstiness separately along the first K principal semantic directions.

This construction supports a positive, component-resolved question: how is semantic burstiness distributed across the principal directions of a text? The principal components are ordered by variance, but PCA does not maximise burstiness. The empirical relation between the eigenvalue spectrum $\{\lambda_k\}$ and the burstiness spectrum $\{B_k\}$ therefore reveals whether directions of strong semantic variation are also directions of strong temporal clustering. The analyses below show interpretable semantic contrasts along the leading components, broad IET distributions in the original texts, and elevated burstiness throughout the first 20 components examined. They also show that the strength of association between semantic variance and burstiness is text dependent: the two quantities are related in some texts but are not interchangeable.

We then use two complementary null models to determine which statistical properties produce this behaviour. The word-shuffled null preserves the multiset of embedded tokens and therefore preserves the empirical covariance, PCA eigensystem, and marginal projected distributions exactly, while destroying temporal order. The FGN surrogate retains the empirical rank-frequency structure and is tuned to match the original rank-sequence DFA exponent, but generates that scaling through a stationary rank-level process. The comparison separates static semantic geometry from ordered semantic dynamics, and global rank-level memory from burstiness in embedding space.

Across the analysed corpus, both controls remain close to the geometric IET baseline, whereas the original text is more bursty across essentially all text–component pairs and over a broad range of threshold quantiles. The projected autocorrelation functions and DFA exponents reinforce this separation: semantic persistence remains visible in the original trajectory but disappears from both controls, including the FGN surrogate whose rank sequence retains the original global DFA exponent. Thus matching lexical frequencies and rank-level long-range scaling is not sufficient to reproduce semantic burstiness.

1.4 Structure of the paper

Section 2 develops the theoretical framework, including semantic projections, PCA, the component-resolved burstiness spectrum, and the two null models. Section 3 describes the computational pipeline. Section 4 presents the Results for four representative texts,

organised around semantic interpretability, component-resolved burstiness, and threshold robustness. Section 5 establishes the corpus-level generality of the component spectra and null-model separation. Section 6 examines persistence in semantic projection space using autocorrelation and DFA and gives an information-theoretic interpretation. Section 7 discusses implications and limitations.

2 Theoretical Framework

2.1 From binary indicators to continuous semantic projections

Following Altmann et al. (2012), let a text be a symbolic sequence $\mathbf{s} = (s_1, s_2, \dots, s_N)$ over a finite vocabulary $\mathcal{A}$. A local matching condition α defines a binary observable

$$u_t^{(\alpha)} = \mathbf{1}[s_t \text{ satisfies } \alpha]. \quad (2)$$

The autocorrelation of this sequence and the diffusion exponent γ , defined by $\sigma_X^2(t) \sim t^\gamma$ for $X(t) = \sum_{j=1}^t u_j^{(\alpha)}$, characterise its long-range structure.

We generalise the local lexical condition to a distributed semantic observable. Let $\phi : \mathcal{A} \rightarrow \mathbb{R}^d$ be a pre-trained static word embedding. After preprocessing and embedding lookup, the retained tokens are relabeled consecutively as $(\tilde{s}_1, \dots, \tilde{s}_M)$, with $M \leq N$. Their *embedding trajectory* is

$$\mathcal{E} = (\mathbf{e}_t)_{t=1}^M, \quad \mathbf{e}_t = \phi(\tilde{s}_t). \quad (3)$$

For a unit vector $\mathbf{v} \in \mathbb{R}^d$, the corresponding *semantic projection* is

$$x_t^{\mathbf{v}} = (\mathbf{e}_t - \bar{\mathbf{e}}) \cdot \mathbf{v}, \quad \bar{\mathbf{e}} = \frac{1}{M} \sum_{t=1}^M \mathbf{e}_t. \quad (4)$$

At threshold θ , this continuous series defines the binary event sequence

$$b_t^{\mathbf{v}}(\theta) = \mathbf{1}[x_t^{\mathbf{v}} > \theta]. \quad (5)$$

The projection describes motion along a semantic direction, while the binary series records visits to its upper tail and supplies the events whose recurrence statistics are analysed.

Remark 1. *If $\mathbf{v}$ is parallel to $\phi(w) - \bar{\mathbf{e}}$ for a word type w , then $x_t^{\mathbf{v}}$ tends to be large when the current token is w and for tokens whose embeddings are close to, or strongly aligned with, $\phi(w)$. Thresholding this projection therefore gives a softened semantic analogue of the exact word indicator in Eq. (2). The construction extends the unit of analysis from one lexical item to a distributed semantic neighbourhood.*

2.2 PCA of the embedding trajectory

The direction $\mathbf{v}$ determines the semantic observable. We use the principal components of the embedding trajectory to obtain a text-specific orthonormal basis ordered by captured variance. The empirical covariance matrix of the mean-centred trajectory is

$$\Sigma = \frac{1}{M} \sum_{t=1}^M (\mathbf{e}_t - \bar{\mathbf{e}})(\mathbf{e}_t - \bar{\mathbf{e}})^\top \in \mathbb{R}^{d \times d}. \quad (6)$$

Its eigendecomposition

$$\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0, \quad (7)$$

yields the principal directions $\mathbf{v}_k$ and their variances λ_k . The projected series for component k is

$$x_t^{(k)} = (\mathbf{e}_t - \bar{\mathbf{e}}) \cdot \mathbf{v}_k, \quad t = 1, \dots, M. \quad (8)$$

PCA directions maximise variance, not semantic interpretability or burstiness. Their semantic meaning is assessed empirically from the word types at the positive and negative poles of each component. The ordered eigenvalue spectrum describes the static distribution of semantic variance, while the temporal statistics of $x_t^{(k)}$ describe how the text moves along these directions.

A key property of Σ is its invariance under permutation of the token sequence. For a permutation π of $\{1, \dots, M\}$,

$$\Sigma^\pi = \frac{1}{M} \sum_{t=1}^M (\mathbf{e}_{\pi(t)} - \bar{\mathbf{e}})(\mathbf{e}_{\pi(t)} - \bar{\mathbf{e}})^\top = \Sigma. \quad (9)$$

It follows that a word-shuffled surrogate has the same eigenvalues, eigenvectors, and marginal distribution of every projected series as the original. PCA therefore defines a static geometry of the embedded lexical composition, whereas burstiness, autocorrelation, and DFA probe the ordering of visits through that geometry.

2.3 The burstiness spectrum

For component k , the projected series is thresholded at its empirical q -quantile:

$$b_t^{(k)} = \mathbf{1} \left[x_t^{(k)} > Q_q \left(\{x_t^{(k)}\}_{t=1}^M \right) \right]. \quad (10)$$

The event rate is $p = 1 - q$ by construction. Let $\tau_1^{(k)}, \tau_2^{(k)}, \dots$ be the gaps between successive positions for which $b_t^{(k)} = 1$. The burstiness parameter is the coefficient of variation of the IET sequence (Goh and Barabási, 2008; Altmann et al., 2012):

$$B_k = \text{CV}(\tau^{(k)}) = \frac{\sigma_\tau^{(k)}}{\langle \tau \rangle^{(k)}}. \quad (11)$$

The collection $\{B_k\}_{k=1}^K$ is the *burstiness spectrum* of the text.

At finite event rate, the appropriate independent-event reference is a geometric distribution rather than an exponential distribution. For an i.i.d. Bernoulli(p) event sequence,

$$P(\tau = n) = (1 - p)^{n-1} p, \quad n \geq 1, \quad (12)$$

which gives

$$\text{CV}_{\text{geom}}(p) = \sqrt{1 - p}. \quad (13)$$

For the default $q = 0.95$, $p = 0.05$ and $\text{CV}_{\text{geom}} = \sqrt{0.95} \simeq 0.975$. Values of B_k above this reference indicate more irregular, clustered recurrence than expected from independent events at the same rate.

All comparisons use the same quantile convention for the original, word-shuffled, and FGN projected series, so the geometric reference is common to the three sequences. For the word-shuffled surrogate, the equality of projected marginal distributions is exact by Eq. (9); implementation details for the FGN construction are given in Section 3.4.

2.4 Semantic variance and burstiness across components

The PCA index k orders directions by decreasing variance, whereas B_k orders no quantity by construction. The pair of spectra

$$\{\lambda_k\}_{k=1}^K, \quad \{B_k\}_{k=1}^K \quad (14)$$

therefore separates two properties of the text: the amount of semantic variation along a direction and the temporal clustering of visits along that direction. The component-resolved analysis asks whether burstiness is elevated above the geometric and surrogate baselines, how it varies across the retained PCA directions, and how strongly it is associated with semantic variance.

We quantify the last relation by the Pearson correlation

$$r(\lambda, B) = \frac{\sum_{k=1}^K (\lambda_k - \bar{\lambda})(B_k - \bar{B})}{\sqrt{\sum_{k=1}^K (\lambda_k - \bar{\lambda})^2} \sqrt{\sum_{k=1}^K (B_k - \bar{B})^2}}, \quad (15)$$

where $\bar{\lambda}$ and $\bar{B}$ denote component means over the retained range. A positive value indicates that high-variance directions tend also to be more bursty, but even a positive correlation does not identify the two spectra: a lower-variance component can be highly bursty if it is associated with a topic concentrated in a limited number of textual regions. The analyses below retain $K = 20$ components and report both the full burstiness spectrum and its relation to the eigenvalue spectrum.

2.5 The word-shuffled null model

The word-shuffled null tests whether elevated burstiness can be explained by static lexical composition and embedding geometry alone. Given the retained token sequence $(\tilde{s}_1, \dots, \tilde{s}_M)$, a surrogate is formed by a uniformly random permutation $\pi \in S_M$. Its embedding trajectory is therefore a permutation of the original trajectory.

By Eq. (9), the surrogate preserves exactly the word counts of the retained sequence, the embedding covariance matrix and complete PCA eigensystem, and the marginal distribution of every projected series $x_t^{(k)}$. It destroys temporal order. Consequently, differences in IET distributions, burstiness, autocorrelation, or DFA arise from the ordering of visits through semantic space rather than from vocabulary, frequency distribution, or static embedding geometry.

2.6 The FGN surrogate as a long-range null model

The FGN surrogate of Degli Esposti and Montemurro (2025) provides a complementary null model. Starting from the Zipf-rank sequence $\mathbf{r} = (r_1, \dots, r_N)$, a fractional Gaussian noise process is mapped onto the empirical rank-frequency structure. Its parameter is chosen by bisection so that the DFA exponent of the surrogate rank sequence matches that of the original:

$$\alpha_{\text{FGN}} \simeq \alpha_{\text{orig}}. \quad (16)$$

The resulting ranks are mapped back to words and processed through the same filtering, embedding, and projection pipeline as the original.

The construction preserves the text length and empirical frequency structure and reproduces the global rank-sequence DFA exponent by design. Unlike the word-shuffled

null, its defining constraint is long-range rank-level scaling rather than exact permutation of the retained embedding trajectory. It therefore tests whether stationary correlations between word ranks are sufficient to produce the burstiness observed along semantic projections.

Remark 2. *The FGN construction does not impose covariance invariance as an algebraic constraint in the way that word shuffling does. In the analysed data, however, the leading FGN and original eigenvalue spectra and directions are empirically very close, as illustrated by the spectrum comparisons in the Results. The surrogates are therefore compared along the original PCA directions, while the word-shuffled case remains the exact geometry-preserving control.*

Remark 3. *Rank proximity is not semantic proximity. Words of similar frequency can occupy widely separated regions of embedding space, so correlations imposed on the rank sequence need not generate correlations between the corresponding embedding vectors. The FGN surrogate tests this distinction directly: it retains global rank-level scaling without explicitly organising semantically related words into the same textual regions.*

2.7 Logical role of the null models

The original, shuffled, and FGN sequences isolate different statistical ingredients. The original contains its lexical composition, static embedding geometry, temporal ordering through that geometry, and observed rank-level scaling. The word-shuffled sequence preserves the first two exactly while removing temporal order. The FGN surrogate preserves the frequency structure and rank-level DFA scaling while imposing no semantic organisation beyond that inherited from the rank-to-word mapping.

The comparisons therefore address two distinct questions. First, does the ordered original trajectory produce component-level burstiness that is absent when the same embedded tokens are randomly permuted? Second, is matching the global DFA exponent of the rank sequence sufficient to reproduce that semantic burstiness? The Results answer both questions by comparing IET distributions, burstiness spectra, autocorrelation functions, and projection-level DFA exponents across the three sequences.

3 Computational Pipeline

The following sections describe the implementation of the framework introduced in Section 2.

3.1 Text preprocessing

Input texts are obtained in plain-text format from Project Gutenberg. Boilerplate headers and footers are stripped by searching for the standard markers `*** START OF` and `*** END OF`. Tokenisation uses the NLTK `word_tokenize` function applied to the lowercased text, retaining only alphabetic tokens (no punctuation, no numerals, no contractions).

Two frequency-based filters are then applied. Let $\text{rank}(w)$ denote the rank of word w in the vocabulary ordered by decreasing frequency.

1. *High-frequency removal*: words with $\text{rank}(w) \leq R$ are removed. Default: $R = 100$.
2. *Hapax removal*: words with frequency $f(w) < F_{\min}$ are removed. Default: $F_{\min} = 5$.

The motivation for the high-frequency filter deserves emphasis. Without it, the dominant PCA direction $\mathbf{v}_1$ of the embedding trajectory captures a *frequency contrast* rather than a topical one: the negative pole concentrates high-frequency function words (*the, of, and, to, a, ...*), and the positive pole concentrates rare content words whose embedding vectors are geometrically distant from the high-frequency cluster. After removing the top- R most frequent words, $\mathbf{v}_1$ captures the dominant *topical* contrast of the text.

The choice $R = 100$ is empirically motivated by direct inspection of word scores along PC1: with $R = 100$, function words are removed from the vocabulary and the principal components reflect genuine topical directions. We verified that increasing R to 150 or 200 does not change the qualitative results but begins to degrade the alignment between the original and FGN covariance structures for higher components (Remark 2).

A known residual artefact persists in PC1 for texts whose vocabulary contains many rare transliterated proper nouns or archaic forms (e.g. *War and Peace*, the King James Bible): these words cluster in a peripheral region of the GloVe embedding space and partially contaminate PC1 regardless of R . This artefact is independent of topical structure and is documented where relevant. Component spectra retain PC1 for completeness, but its semantic interpretation and short-lag temporal statistics are treated cautiously; results for $k \geq 2$ are unaffected by this particular artefact.

3.2 Word embedding model

We use GloVe vectors trained on Wikipedia and Gigaword (Pennington et al., 2014), with $d = 300$ dimensions and a vocabulary of approximately 400,000 words. The implementation uses the `glove-wiki-gigaword-300` model, loaded through `gensim.downloader`.

The choice of static (non-contextual) embeddings is deliberate. For static embeddings, $\mathbf{e}_t = \phi(s_t)$ depends only on the word *type* s_t , making the trajectory a deterministic function of the token sequence. Shuffling the token sequence thus has an unambiguous effect: it permutes the trajectory while leaving the multiset $\{\mathbf{e}_t\}$ unchanged. Contextual embeddings (BERT, GPT) would assign different vectors to the same word type depending on local context, introducing dependencies between the embedding and the ordering that would complicate the surrogate comparison. We defer contextual embedding analysis to future work (Section 7).

Out-of-vocabulary tokens are skipped; their original positions are not used in the trajectory or in IET computations. The OOV rate after filtering is typically 5–15% for Victorian English texts, consisting mainly of proper nouns, archaic forms, and dialect words.

3.3 PCA and projection

The embedding trajectory $(e_1, \dots, e_M) \subset \mathbb{R}^d$ is assembled as a matrix $\mathbf{V} \in \mathbb{R}^{M \times d}$. After mean-centring, $\mathbf{X} = \mathbf{V} - \mathbf{1}\bar{\mathbf{e}}^\top$, PCA is computed via `sklearn.decomposition.PCA` retaining $K_{\text{show}} = 50$ components for eigenvalue analysis and $K_{\text{burst}} = 20$ for the burstiness spectrum. The projected series matrix $\mathbf{P} \in \mathbb{R}^{M \times K_{\text{show}}}$ with $P_{tk} = x_t^{(k)}$ is used for all subsequent analyses.

3.4 FGN surrogate generation

The FGN surrogate is constructed using the pipeline of Degli Esposti and Montemurro (2025):

1. The text is tokenised and each token is replaced by its Zipf rank, producing a rank sequence $\mathbf{r}$.
2. The DFA exponent α of the original text is estimated from $\mathbf{r}$.
3. A FGN process with parameter α_0 is generated and mapped onto the empirical rank histogram. The parameter α_0 is found by bisection so that the DFA exponent of the surrogate matches α .
4. The rank sequence of the surrogate is mapped back to words via the inverse Zipf mapping, producing a word sequence $\tilde{\mathbf{s}}$.

The resulting word sequence is then processed through the same frequency filter (Section 3.1), embedding lookup, and projection pipeline as the original text. The surrogate is projected onto the *original* PCA eigenvectors $\{\mathbf{v}_k\}$ to ensure comparability along identical semantic directions. Because $\Sigma_{\text{FGN}} \approx \Sigma_{\text{orig}}$ (Remark 2), the original eigenvectors are a near-optimal basis for the FGN trajectory as well; any residual difference is below 3% in eigenvalue and below 5% in eigenvector angle for $k \leq 20$ at $R = 100$.

4 Results: semantic directions and burstiness spectra

4.1 Representative texts and summary

We first examine four texts chosen to span distinct forms of semantic and narrative organisation: a long polyphonic novel (*War and Peace*), a scientific treatise (*On the Origin of Species*), a Victorian novel of formation (*Great Expectations*), and a modernist novel with marked episode-to-episode stylistic variation (*Ulysses*). All analyses use the pipeline of Section 3, with $R = 100$, $F_{\min} = 5$, $q = 0.95$, and $K = 20$ components unless stated otherwise.

The Results are organised by empirical question rather than by book. We first establish that the PCA basis defines interpretable, text-specific directions while remaining a static property of lexical composition. We then examine the inter-event-time distributions and the component-resolved burstiness spectra, followed by robustness to the threshold quantile. Table 1 collects the main numerical descriptors.

Table 1: Summary statistics for the four single-text analyses. N : raw tokens; M : filtered tokens ($R=100$, $F_{\min} = 5$); OOV: out-of-vocabulary rate after filtering; α : DFA exponent; α_0 : FGN bisection parameter (resolved to ± 0.01 by the bisection tolerance); $r(\lambda, B)$: Pearson correlation between eigenvalue and burstiness spectra; B_2 : burstiness of PC2 at $q = 0.95$.

Text	N	M	OOV	α	α_0	$r(\lambda, B)$	B_2
War and Peace	564,788	235,933	1.6%	0.721	0.76	0.702***	1.855
Origin of Species	151,179	58,913	0.4%	0.771	0.80	0.295	1.888
Great Expectations	188,925	64,410	1.3%	0.689	0.74	0.511*	1.209
Ulysses	264,165	101,489	0.7%	0.763	0.80	0.571**	1.495

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; no mark: $p > 0.05$.

Across the four texts, the original projection sequences are more bursty than both controls for all $K = 20$ components at $q = 0.95$. The correlation $r(\lambda_k, B_k)$ varies from 0.294

to 0.702, showing that captured semantic variance and temporal clustering are neither independent in every text nor interchangeable.

4.2 Static semantic geometry and interpretable directions

The eigenvalue spectra describe the geometry of the retained embedded vocabulary. As established in Section 2.2, word shuffling leaves this geometry exactly invariant. The FGN spectra are also empirically close to the originals over the displayed range, although that agreement is not imposed algebraically. The semantic poles of the leading components provide a separate empirical test of whether those directions correspond to interpretable contrasts.

4.2.1 *War and Peace*

War and Peace is the longest representative text ($N = 564,788$ raw tokens). After filtering and embedding lookup, the trajectory contains $M = 235,933$ tokens; the rank-sequence DFA exponent is $\alpha = 0.721$, and the fitted FGN parameter is $\alpha_0 = 0.76$. Figure 1 shows that PC1 and PC2 account for 6.98% and 4.21% of the variance, respectively, and the first ten components account for 27.2%.

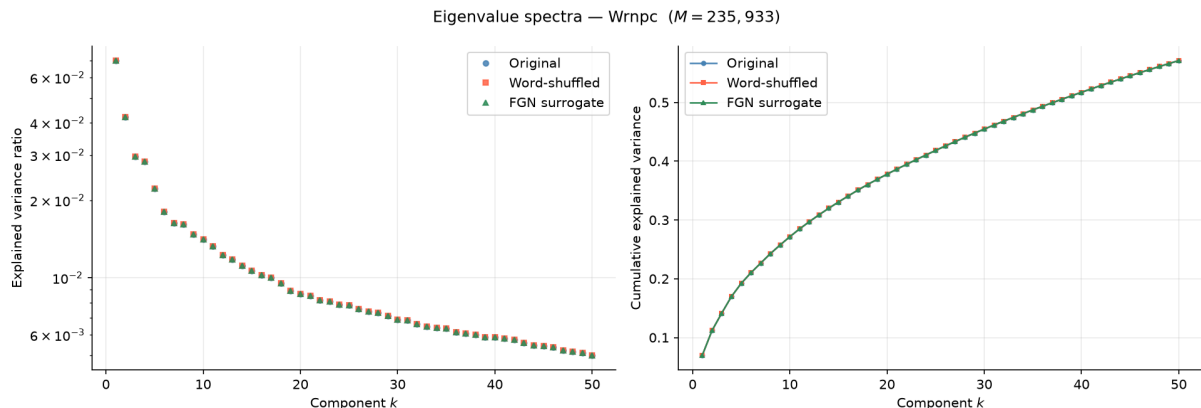


Figure 1: Eigenvalue spectra and cumulative explained variance for *War and Peace*: original (blue), word-shuffled (red), and FGN surrogate (green). The shuffled spectrum is identical to the original by permutation invariance; the FGN spectrum is empirically very close over the displayed range (Remark 2).

The semantic poles in Table 2 reveal two especially clear contrasts. PC2 separates military vocabulary (*regiment, infantry, cavalry, battalion*) from personal and emotional discourse (*really, yourself, feel, understand*). PC3 contrasts physical action and embodied presence with philosophical and moral reflection. PC1 partly reflects the known GloVe artefact in which transliterated Russian names and emotionally inflected rare words lie at the periphery of the embedding cloud; it is retained in the spectra but interpreted cautiously.

4.2.2 *On the Origin of Species*

On the Origin of Species contains $N = 151,179$ raw tokens and $M = 58,913$ retained tokens, with $\alpha = 0.771$ and $\alpha_0 = 0.80$. PC1 explains 6.61% of the variance, PC2 3.58%,

Table 2: Top-10 word types at the positive and negative poles of PC1, PC2, and PC3 for *War and Peace* ($R = 100$, $F_{\min} = 5$). Frequencies in the filtered token sequence.

PC	Pole	Top words (freq)	Interpretation
PC1	+	<i>gerasim</i> (26), <i>pavlovna</i> (151), <i>petya</i> (268), <i>dmitrievna</i> (105), <i>piteously</i> (7), <i>irritably</i> (6)	Russian proper names + emotional adverbs (GloVe artefact)
PC1	-	<i>any</i> (465), <i>government</i> (63), <i>people</i> (557), <i>can</i> (631), <i>our</i> (619), <i>country</i> (106)	High-frequency function/abstract English (GloVe artefact)
PC2	+	<i>regiment</i> (226), <i>infantry</i> (82), <i>cavalry</i> (77), <i>battalion</i> (44), <i>squadron</i> (64), <i>army</i> (663)	Military narrative (War)
PC2	-	<i>really</i> (203), <i>yourself</i> (117), <i>feel</i> (133), <i>myself</i> (148), <i>think</i> (357), <i>understand</i> (347)	Personal/emotional discourse (Peace)
PC3	+	<i>walked</i> (70), <i>sat</i> (345), <i>feet</i> (112), <i>screaming</i> (6), <i>legs</i> (101), <i>bed</i> (118)	Physical action, embodied presence
PC3	-	<i>principles</i> (7), <i>obligations</i> (6), <i>moral</i> (32), <i>justification</i> (11), <i>renounce</i> (8), <i>unjust</i> (7)	Philosophical/moral reflection

and the first ten components 26.4%. The original, shuffled, and FGN spectra are again closely aligned (Figure 15).

PC1 is dominated by specialised nineteenth-century biological and geological terminology at one pole and pronouns or conversational markers at the other, and therefore contains a register-related embedding artefact. PC2 is a substantive geographical-versus-epistemic direction: spatial and biogeographical terms are opposed to the language of inference and theoretical reasoning. PC3 contrasts first-person field observations and concrete specimens with abstract classification and synthesis. These poles are listed in Table 3.

4.2.3 *Great Expectations*

Great Expectations contains $N = 188,925$ raw tokens and $M = 64,410$ retained tokens, with $\alpha = 0.689$ and $\alpha_0 = 0.74$. PC1, PC2, and the first ten components account for 6.54%, 4.34%, and 27.2% of the variance, respectively. The three covariance spectra are again nearly coincident (Figure 16).

The positive pole of PC1 contains character names and Dickensian dialect, opposed to generic vocabulary, and is treated as an embedding artefact. PC2 contrasts spatial and kinetic language associated with the river, marshes, and forge with introspective and conversational language. PC3 opposes embodied physical description to legal and institutional vocabulary, reflecting the tension between Pip’s lived world and the machinery of inheritance and social advancement. The corresponding poles are listed in Table 4.

4.2.4 *Ulysses*

Ulysses contains $N = 264,165$ raw tokens and $M = 101,489$ retained tokens, with $\alpha = 0.763$ and $\alpha_0 = 0.80$. Its first ten components account for 25.0% of the variance, the smallest

Table 3: Top-10 word types at the positive and negative poles of PC1, PC2, and PC3 for *On the Origin of Species* ($R = 100$, $F_{\min} = 5$).

PC	Pole	Top words (freq)	Interpretation
PC1	+	<i>sessile</i> (11), <i>faunas</i> (14), <i>ovules</i> (6), <i>polymorphic</i> (9), <i>insectivorous</i> (6), <i>larvae</i> (26), <i>palaeozoic</i> (15)	Specialised 19th-c. natural history lexicon
PC1	-	<i>he</i> (163), <i>him</i> (17), <i>said</i> (47), <i>us</i> (104), <i>my</i> (150), <i>who</i> (75)	Pronouns / interpersonal register (GloVe artefact)
PC2	+	<i>northern</i> (37), <i>islands</i> (131), <i>sea</i> (100), <i>southern</i> (43), <i>tropical</i> (16), <i>archipelago</i> (37), <i>coast</i> (18)	Geographical distribution chapters
PC2	-	<i>understand</i> (74), <i>infer</i> (31), <i>necessarily</i> (27), <i>plainly</i> (43), <i>suppose</i> (63), <i>ought</i> (40)	Scientific reasoning and theoretical inference
PC3	+	<i>beaks</i> (10), <i>ants</i> (27), <i>legs</i> (37), <i>dog</i> (20), <i>flowers</i> (61), <i>beetles</i> (17), <i>me</i> (153)	Field observations, specific specimens, first-person
PC3	-	<i>geographical</i> (42), <i>distribution</i> (55), <i>proportional</i> (16), <i>classification</i> (49), <i>development</i> (28), <i>fundamental</i> (9)	Abstract synthesis, quantitative generalisation

Table 4: Top-10 word types at the positive and negative poles of PC1, PC2, and PC3 for *Great Expectations* ($R = 100$, $F_{\min} = 5$).

PC	Pole	Top words (freq)	Interpretation
PC1	+	<i>jiggers</i> (242), <i>estella</i> (270), <i>biddy</i> (231), <i>provis</i> (70), <i>magwitch</i> (26), <i>limekiln</i> (6), <i>tickler</i> (10)	Character names + Victorian dialect (GloVe artefact)
PC1	-	<i>will</i> (174), <i>any</i> (268), <i>their</i> (175), <i>two</i> (254), <i>last</i> (185), <i>other</i> (201)	High-frequency generic vocabulary
PC2	+	<i>river</i> (59), <i>near</i> (66), <i>opened</i> (41), <i>fell</i> (55), <i>port</i> (7), <i>yards</i> (9), <i>miles</i> (17)	Spatial/kinetic (Thames, marshes, forge)
PC2	-	<i>really</i> (41), <i>think</i> (241), <i>myself</i> (235), <i>understand</i> (53), <i>feel</i> (28), <i>ashamed</i> (15), <i>want</i> (93)	Introspective/conversational register
PC3	+	<i>legs</i> (35), <i>walked</i> (45), <i>feet</i> (34), <i>knees</i> (15), <i>fingers</i> (31), <i>shoulder</i> (54), <i>bed</i> (74)	Physical, embodied description
PC3	-	<i>authority</i> (8), <i>constitutional</i> (5), <i>law</i> (9), <i>appointed</i> (11), <i>established</i> (13), <i>derived</i> (13)	Legal/institutional machinery

fraction among the four texts, consistent with the diversity of registers across its eighteen episodes (Figure 17).

PC1 is strongly affected by Dublin-specific proper names, multilingual fragments, and neologisms. PC2 provides a more substantive formal-public versus intimate-private contrast, separating titles, institutional names, and parodic catalogue language from colloquial and sensory interior discourse. PC3 contrasts physical-material description with vocabulary of family, religion, and literary identity. These semantic poles are listed in Table 5.

Table 5: Top-10 word types at the positive and negative poles of PC1, PC2, and PC3 for *Ulysses* ($R = 100$, $F_{\min} = 5$).

PC	Pole	Top words (freq)	Interpretation
PC1	+	<i>dedalus</i> (174), <i>dignam</i> (83), <i>lenehan</i> (107), <i>gerty</i> (64), <i>virag</i> (43), <i>voglio</i> (9)	Dublin proper names + multilingual fragments (GloVe artefact)
PC1	−	<i>government</i> (14), <i>people</i> (78), <i>country</i> (54), <i>world</i> (155), <i>united</i> (10), <i>china</i> (9)	Abstract political vocabulary (GloVe artefact)
PC2	+	<i>sir</i> (219), <i>william</i> (40), <i>john</i> (196), <i>royal</i> (47), <i>edward</i> (17), <i>founded</i> (8), <i>lieutenant</i> (13)	Formal titles, institutional names, parodic chapters
PC2	−	<i>yourself</i> (38), <i>feel</i> (104), <i>myself</i> (71), <i>really</i> (49), <i>laugh</i> (37), <i>smell</i> (50), <i>awfully</i> (22)	Intimate register, internal monologue episodes
PC3	+	<i>thick</i> (26), <i>surface</i> (10), <i>mud</i> (6), <i>feet</i> (82), <i>yellow</i> (50), <i>sand</i> (32), <i>shells</i> (14)	Physical-material description (Sandymount Strand, streets)
PC3	−	<i>married</i> (56), <i>son</i> (113), <i>wife</i> (140), <i>brother</i> (66), <i>god</i> (247), <i>dear</i> (83), <i>friend</i> (68)	Family, religion, literary identity (Telemachus, Nostos)

Taken together, the four texts show that PCA produces an ordered static geometry whose leading directions are often semantically interpretable, but whose first component can be contaminated by peripheral lexical clusters. The temporal question is separate: whether visits along these directions occur regularly or in concentrated textual episodes.

4.3 Burstiness across principal components

We now turn from static geometry to temporal organisation. For each component, threshold crossings define event sequences and IET distributions. Figure 3 and Figures 2, 18, 19 and 20 show the representative-text evidence. In every text, the original series has broader IET distributions and larger B_k than either control. The spectra are component dependent and generally largest among the early PCs, but the decay with component rank is not monotone. This non-monotonicity is expected because PCA orders variance rather than burstiness.

4.3.1 *War and Peace*

For PC1–PC3, the original IET distributions extend well beyond the geometric reference, whereas the shuffled and FGN controls closely follow it (Figure 2). The burstiness spectrum in Figure 3A is above both controls for all 20 components. PC3 has the largest value, $B_3 = 2.045$, approximately 2.1 times the geometric baseline and more than twice the corresponding null values ($B_3^{\text{surr}} = 0.976$, $B_3^{\text{FGN}} = 0.974$). This is consistent with the concentration of philosophical-reflection passages in particular parts of the novel. The eigenvalue–burstiness correlation is positive but imperfect, $r = 0.702$ ($p = 0.0006$).

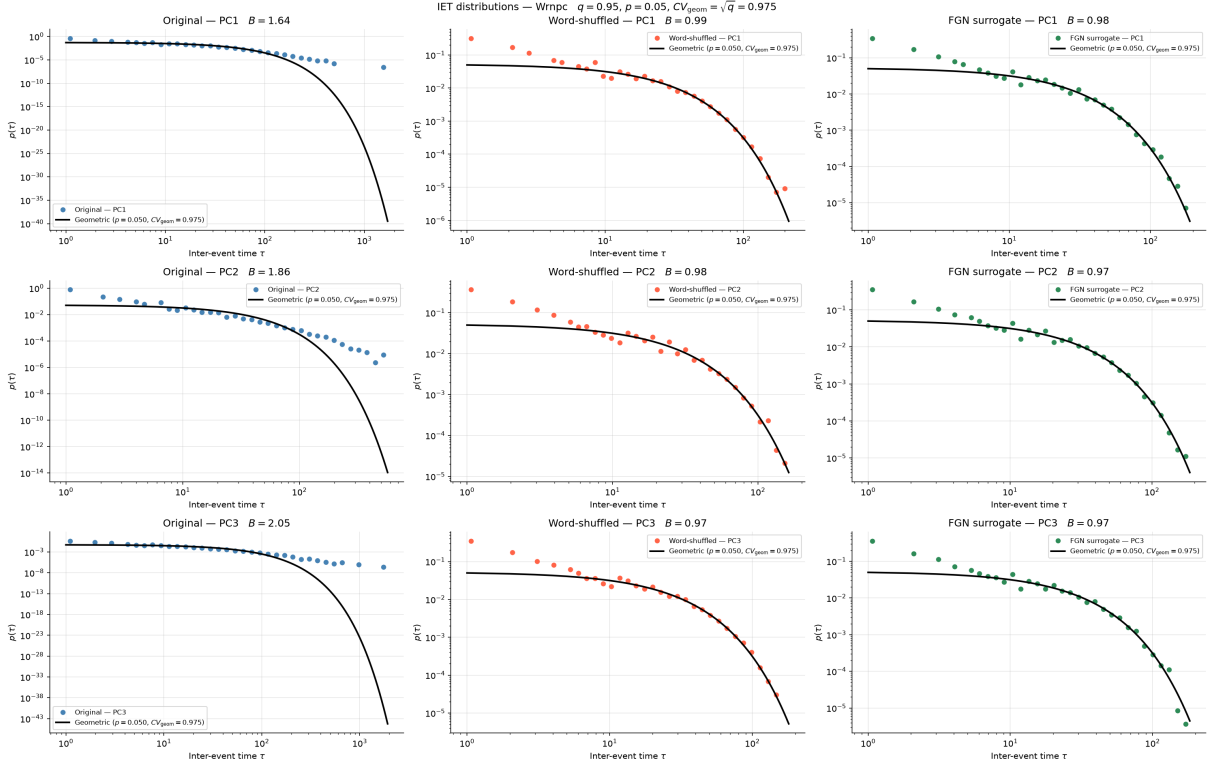


Figure 2: Inter-event-time distributions for PC1–PC3 of *War and Peace* at $q = 0.95$. Columns show the original text (blue), word-shuffled null (red), and FGN surrogate (green). The original projections have broad tails, whereas both controls lie close to the geometric reference.

4.3.2 *On the Origin of Species*

The original projections have broad IET tails, particularly for PC2 and PC3, while both controls remain close to geometric (Figure 18). PC2 is the burstiest direction, $B_2 = 1.888$, followed by PC3 with $B_3 = 1.727$; PC1 is lower at $B_1 = 1.255$. The full original-over-control ordering holds for all 20 components. The weak eigenvalue–burstiness correlation, $r = 0.295$ ($p = 0.207$), shows especially clearly that the direction of greatest semantic variance need not be the direction of greatest temporal clustering.

4.3.3 *Great Expectations*

The original PC1–PC3 IET distributions again have broader tails than the controls (Figure 19). At $q = 0.95$, $B_1 = 1.228$, $B_2 = 1.209$, and $B_3 = 1.244$, while the null values lie

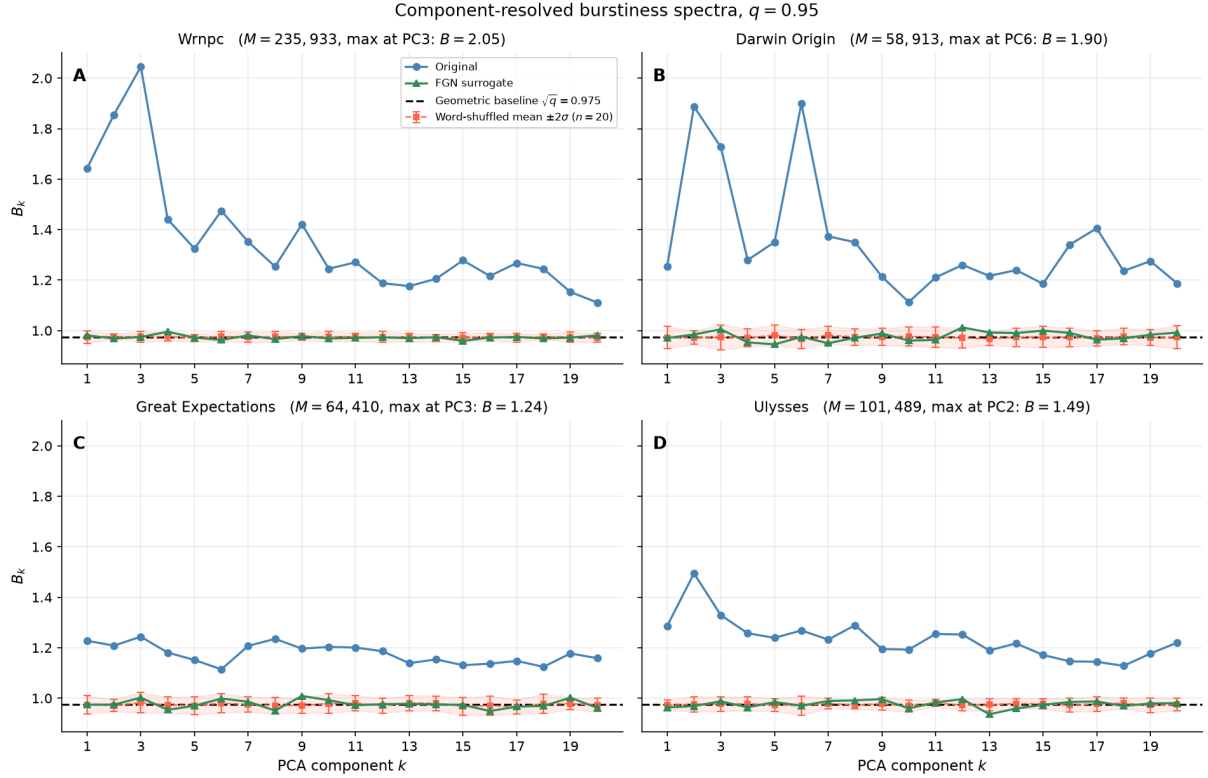


Figure 3: Component-resolved burstiness spectra at $q = 0.95$ for the four representative texts: **A** *War and Peace*, **B** *On the Origin of Species*, **C** *Great Expectations*, **D** *Ulysses*. In every panel the original (blue circles) lies above the FGN surrogate (green triangles), the word-shuffled mean $\pm 2\sigma$ band (red), and the geometric baseline $\sqrt{q} \simeq 0.975$, for all 20 components. The position of the maximum differs between texts (PC3, PC6, PC3 and PC2 respectively) and burstiness is not monotone in PCA rank, because PCA orders components by variance rather than by temporal clustering.

in 0.95–1.00. The original remains above both controls over all 20 components (Figure 3C); the spectrum is flatter and lower overall than for *War and Peace*. The correlation between eigenvalue and burstiness is $r = 0.511$ ($p = 0.021$).

4.3.4 *Ulysses*

For *Ulysses*, the original IET distributions are broad for all three displayed components, while both controls collapse towards the geometric reference (Figure 20). The spectrum peaks at PC2, $B_2 = 1.495$, with $B_1 = 1.290$ and $B_3 = 1.329$ (Figure 3D). The original exceeds both controls throughout the first 20 components, and the eigenvalue–burstiness correlation is $r = 0.571$ ($p = 0.009$).

Averaged over the first 20 components at $q = 0.95$, the representative texts have $\bar{B}^{\text{orig}}$ of 1.36 (*War and Peace*), 1.35 (*On the Origin of Species*), 1.23 (*Ulysses*) and 1.18 (*Great Expectations*), against $\bar{B} \simeq 0.97$ –0.98 for both null models in every case. The ordering is not a simple function of either text length or rank-sequence DFA exponent. The common feature is that component-level burstiness is present in the original semantic trajectories and absent from both null models.

4.4 Robustness to the threshold quantile

The component separation is not tied to the default $q = 0.95$. Figure 4 shows the mean burstiness over $k = 1, \dots, 20$ as q varies from 0.60 to 0.99; the corpus-level version of the ordering fraction is given in Section 5.3. For each of the four representative texts, the ordering holds for all 20 components throughout the displayed range. As expected, the absolute burstiness increases towards more extreme thresholds because rare visits to the upper tail accentuate episode-to-episode clustering.

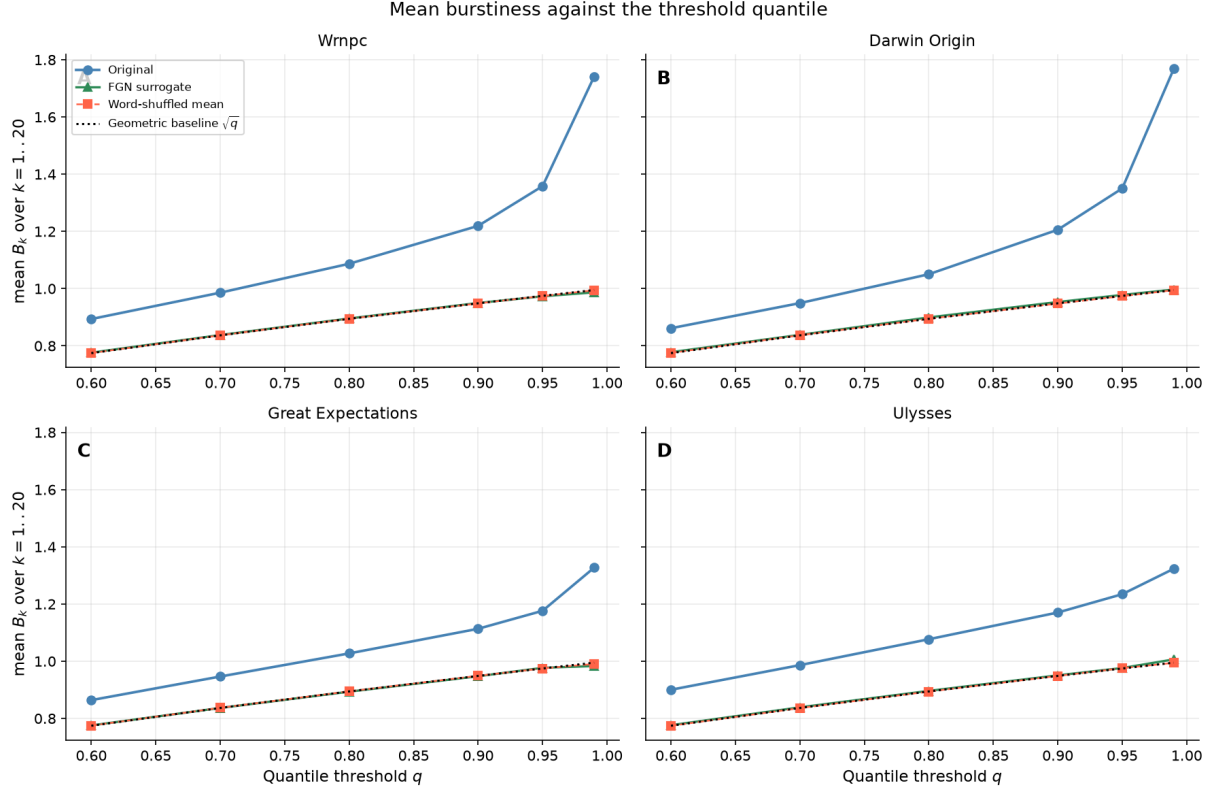


Figure 4: Mean burstiness over $k = 1, \dots, 20$ as a function of the threshold quantile q , for the four representative texts: **A** *War and Peace*, **B** *On the Origin of Species*, **C** *Great Expectations*, **D** *Ulysses*. The original stays above both controls and above the geometric baseline $\sqrt{q}$ throughout $q \in [0.60, 0.99]$, and the full ordering $B_k^{\text{orig}} > B_k^{\text{FGN}}$ and $B_k^{\text{orig}} > B_k^{\text{surr}}$ holds for all 20 components at every displayed q in each text. Absolute burstiness grows towards more extreme thresholds, as rare visits to the upper tail accentuate episode-to-episode clustering.

5 Corpus-level generalisation

Having established the component-resolved pattern in four representative texts (Section 4), we now assess its generality across the full corpus.

5.1 Corpus description

The corpus comprises 16 English literary texts from Project Gutenberg, spanning novels, scientific treatises, and travel writing published between the eighteenth and early twentieth centuries (Table 6). Raw token counts range from approximately 56,000 (*The Picture of Dorian Gray*) to 565,000 (*War and Peace*), with filtered token counts M (after high-frequency and hapax removal, OOV excluded) between 17,000 and 236,000. DFA scaling exponents α lie in the range $[0.63, 0.85]$, consistent with the literature on long-range correlations in English texts (Ebeling and Neiman, 1995; Montemurro and Pury, 2002).

Per-text IET distributions and burstiness spectra for all texts and threshold values can be regenerated with `scripts/run_text.py` in the code repository. The main text reports the aggregated corpus results.

5.2 Component-resolved burstiness holds corpus-wide

The corpus-level component spectrum is summarised in Figure 5. Averaging the burstiness spectrum $\{B_k\}$ across all 16 texts at quantile $q = 0.95$, the original texts are systematically more bursty than both null models across all 20 PCA components: $\langle B_k^{\text{orig}} \rangle_{\text{texts}} \in [1.16, 1.54]$ (mean 1.26), while both null models cluster tightly around the geometric baseline $\text{CV}_{\text{geom}} = \sqrt{0.95} \approx 0.975$: $\langle B_k^{\text{surr}} \rangle \in [0.973, 0.978]$ and $\langle B_k^{\text{FGN}} \rangle \in [0.958, 0.983]$.

At the level of individual (text, k) pairs ($16 \times 20 = 320$ pairs):

- $B_k^{\text{orig}} > B_k^{\text{surr}}$ in **100%** of pairs.
- $B_k^{\text{orig}} > B_k^{\text{FGN}}$ in **100%** of pairs.
- $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$ in **95.6%** of pairs.

The appropriate summary statistic for the magnitude of the effect is the *per-text mean ratio*, computed by averaging $B_k^{\text{orig}}/B_k^{\text{surr}}$ over the 20 components for each text and then averaging over the 16 texts:

$$\left\langle \frac{B_k^{\text{orig}}}{B_k^{\text{surr}}} \right\rangle_{\text{texts}} = 1.29 \pm 0.03, \quad t = 8.7, \quad p < 10^{-6}, \quad (17)$$

where $\pm$ denotes the standard error of the mean over 16 texts. The original texts are thus on average $\sim 30\%$ more bursty than the geometric null. The per-text values range from 1.17 (*Pride and Prejudice*) to 1.39 (*War and Peace*) for fifteen of the sixteen texts, with *Principia Mathematica* as an outlier at 1.73; the spread across texts ($\text{sd} = 0.13$) is itself a signature of how differently topical structure is organised in different works. The analogous ratio for the FGN surrogate is $\langle B_k^{\text{orig}}/B_k^{\text{FGN}} \rangle = 1.29 \pm 0.03$, indistinguishable from the word-shuffled value.

The two null models satisfy $\langle B_k^{\text{FGN}}/B_k^{\text{surr}} \rangle = 1.001 \pm 0.001$ (range over texts $[0.996, 1.005]$; $t = 1.1$, $p = 0.28$): the FGN surrogate and the word-shuffled null are statistically indistinguishable at the corpus level (Section 5.4).

Table 6: Corpus of 16 literary and scientific texts. N : total token count after tokenisation. M : content tokens after frequency filtering (top $R = 100$ words removed, minimum frequency 5). α : DFA scaling exponent of the rank sequence. α_0 : FGN generator exponent matched to α (bisection tolerance 0.01).

Title	Author	N	M	α	α_0
Pride and Prejudice	Austen	122,226	42,364	0.642	0.67
Don Quixote (transl.)	Cervantes	403,152	147,279	0.711	0.76
On the Origin of Species	Darwin	151,179	58,913	0.771	0.80
The Voyage of the Beagle	Darwin	206,728	84,931	0.629	0.67
David Copperfield	Dickens	363,557	134,409	0.727	0.78
Great Expectations	Dickens	188,925	64,410	0.689	0.74
Oliver Twist	Dickens	161,529	59,729	0.711	0.76
Ulysses	Joyce	264,165	101,489	0.763	0.80
The Jungle Book	Kipling	151,161	49,522	0.775	0.83
Moby Dick	Melville	215,652	82,773	0.694	0.73
Principia Mathematica	Newton	112,225	26,347	0.815	0.85
The Analysis of Mind	Russell	88,342	30,449	0.743	0.76
War and Peace	Tolstoy	564,788	235,933	0.721	0.76
Huckleberry Finn	Twain	145,924	52,741	0.774	0.83
Tom Sawyer	Twain	71,102	22,085	0.788	0.80
The Picture of Dorian Gray	Wilde	55,619	16,776	0.845	0.87

5.3 Robustness across quantile threshold

Figure 6 shows the fraction of (text, k) pairs satisfying the ordering conditions as a function of $q \in \{0.60, 0.70, 0.80, 0.90, 0.95, 0.99\}$.

The full ordering $B_k^{\text{orig}} > B_k^{\text{surr}}$ and $B_k^{\text{orig}} > B_k^{\text{FGN}}$ is satisfied in 100% of pairs for all $q \in [0.60, 0.95]$, and in 98.1% of pairs at the extreme $q = 0.99$. The null-model indistinguishability criterion $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$ is met by $\geq 99\%$ of pairs for $q \leq 0.90$, by 96% at $q = 0.95$, and decreases to 73% at $q = 0.99$. This degradation at the extreme threshold is expected: with $p = 0.01$ and $M \approx 80,000$ filtered tokens, there are on average only ~ 800 events per series, giving an IET estimator with relative error $\sim 1/\sqrt{800} \approx 3.5\%$ — comparable to the magnitude of $B^{\text{FGN}} - B^{\text{surr}}$ at this threshold, and larger still for the shortest texts ($M < 30,000$). For $q = 0.95$ ($n_{\text{events}} \approx 4,000$, relative error $\sim 1.6\%$), the estimator is well within the regime where the null-model comparison is reliable.

5.4 Distribution of burstiness ratios

Figure 7 shows the distribution of the ratios $B^{\text{orig}}/B^{\text{surr}}$, $B^{\text{orig}}/B^{\text{FGN}}$, and $B^{\text{FGN}}/B^{\text{surr}}$ over all 320 (text, k) pairs at $q = 0.95$.

Panel A shows that both orig–null ratios are concentrated well above 1, with virtually no mass below 1. The two distributions (orig/surr and orig/FGN) are nearly identical in shape and location, confirming that the FGN and word-shuffled null models are equally poor surrogates for the original burstiness.

Panel B isolates the ratio $B^{\text{FGN}}/B^{\text{surr}}$, which is tightly concentrated around 1 (mean 1.0008, s.e. 0.0012, range $[0.91, 1.09]$). A one-sample t -test against ratio = 1 gives $t = 0.63$, $p = 0.53$: the two null models are statistically indistinguishable, and the distribution is symmetric with no indication of a directional bias.

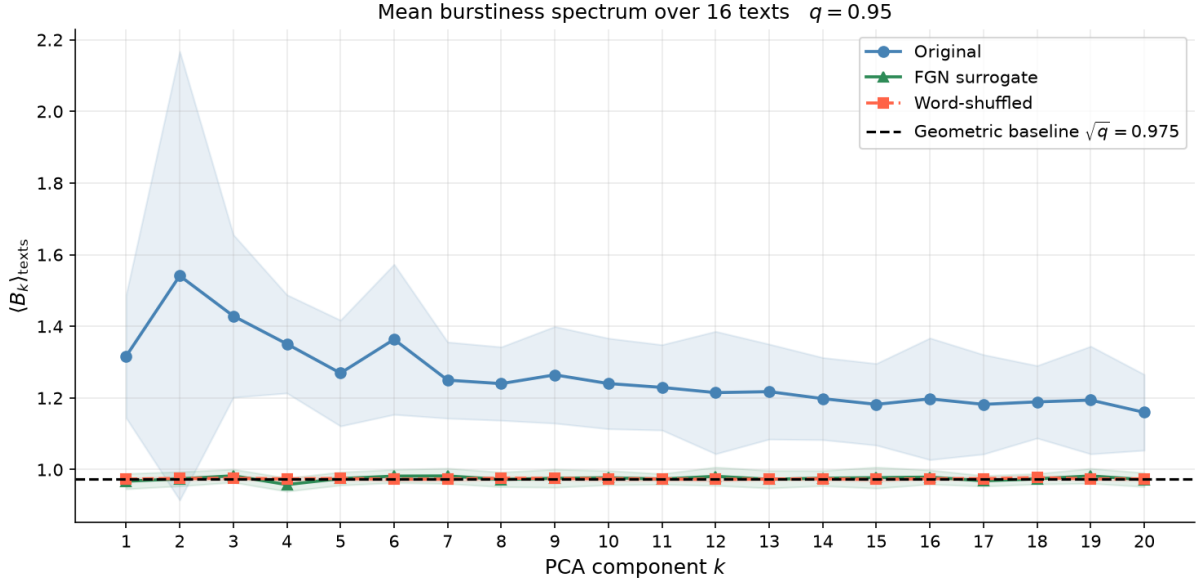


Figure 5: Mean component-resolved burstiness spectrum across the analysed corpus at $q = 0.95$. Curves show the mean $\langle B_k \rangle$ over texts for $k = 1, \dots, 20$; shaded bands indicate one standard deviation. The original texts (blue) are above the FGN surrogate (green), word-shuffled null (red), and geometric baseline $\sqrt{q} \simeq 0.975$ at every component. The largest corpus means occur among the early components, followed by a broad decline with non-monotone secondary peaks.

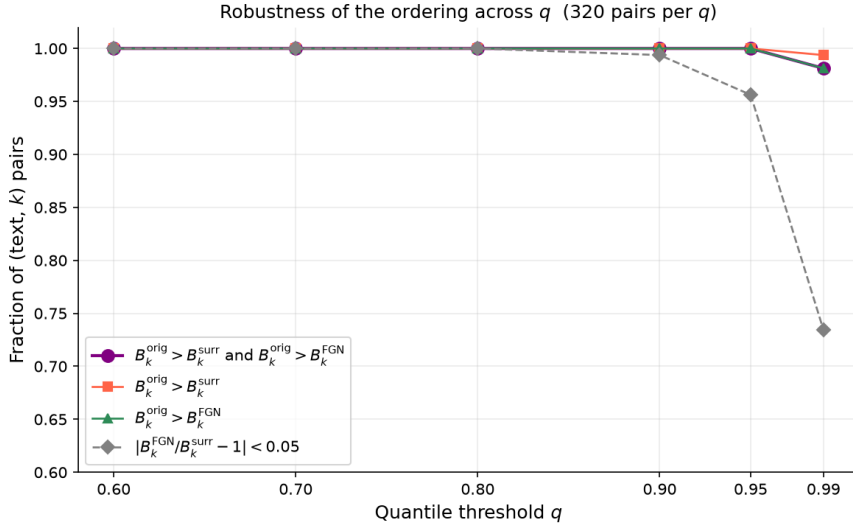


Figure 6: **Robustness of the burstiness ordering across quantile threshold q .** Each point: fraction of (text, k) pairs (320 per q) satisfying the indicated condition. Purple circles: full ordering $B_k^{\text{orig}} > B_k^{\text{surr}}$ and $B_k^{\text{orig}} > B_k^{\text{FGN}}$. Red squares: $B_k^{\text{orig}} > B_k^{\text{surr}}$. Green triangles: $B_k^{\text{orig}} > B_k^{\text{FGN}}$. Grey diamonds: $|B_k^{\text{FGN}}/B_k^{\text{surr}} - 1| < 0.05$. The ordering is robust for $q \in [0.60, 0.95]$; the slight degradation at $q = 0.99$ is due to estimator noise at $p = 0.01$.

Together, Figures 5–7 establish the corpus-level result:

$$B_k^{\text{orig}} \gg B_k^{\text{FGN}} \approx B_k^{\text{surr}} \quad \forall k, \forall \text{text in corpus}, \quad (18)$$

with a $\sim 30\%$ mean excess of the original above the geometric null, independent of text length, genre, DFA exponent, and choice of quantile threshold.

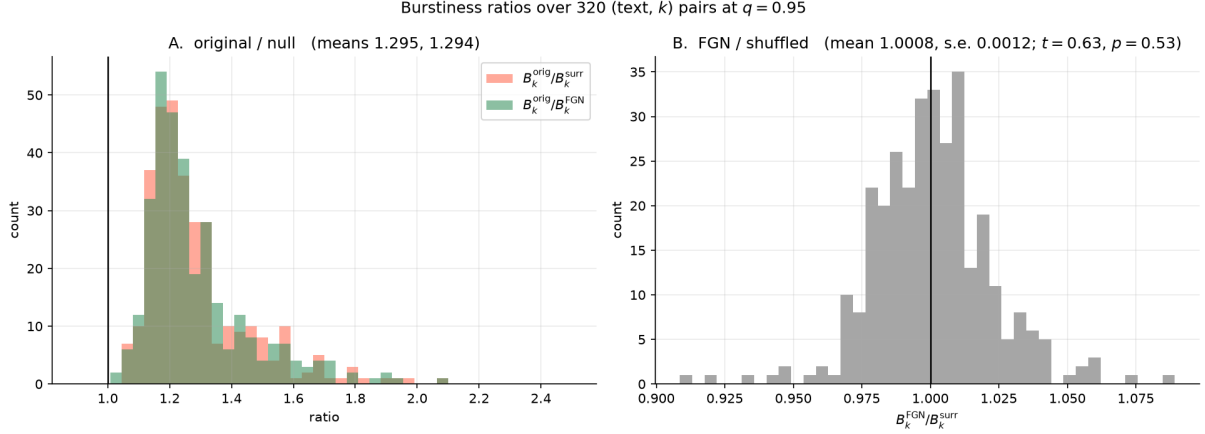


Figure 7: **Distribution of burstiness ratios over 320 (text, k) pairs at $q = 0.95$.** **Panel A:** distributions of $B_k^{\text{orig}}/B_k^{\text{surr}}$ (red) and $B_k^{\text{orig}}/B_k^{\text{FGN}}$ (green), both concentrated well above 1. The two distributions are nearly identical, confirming that the two null models are equivalently far below the original. **Panel B:** distribution of $B_k^{\text{FGN}}/B_k^{\text{surr}}$, tightly centred on 1 (mean 1.0008, s.e. 0.0012; $t = 0.63$, $p = 0.53$), statistically indistinguishable from unity, confirming $B_k^{\text{FGN}} \approx B_k^{\text{surr}}$ across the entire corpus.

6 Temporal persistence in semantic projection space

Sections 4 and 5 establish that the original semantic trajectories have elevated component-resolved burstiness and are separated from both controls. We now test the same temporal organisation in the continuous projections using normalised autocorrelation functions and DFA. An information-theoretic argument explains why the long-range correlations imposed on the FGN rank sequence need not survive the mapping into embedding space.

6.1 Conditional entropy of the embedding trajectory

The qualitative argument of Remark 3 can be cast in information-theoretic terms. Consider the embedding trajectory $\mathcal{E} = (\mathbf{e}_t)_{t=1}^M$ as a stationary (or locally stationary) stochastic process over $\mathbb{R}^d$. For a lag $\ell \geq 1$, define the *conditional entropy*

$$H_\ell = H(\mathbf{e}_t \mid \mathbf{e}_t[t-1], \dots, \mathbf{e}_t[t-\ell]), \quad (19)$$

where $\mathbf{e}_t[s] \equiv \mathbf{e}_s$ for brevity. For an i.i.d. process, $H_\ell = H(\mathbf{e}_t)$ for all $\ell \geq 1$: past embeddings carry no information about the present. For a process with long-range semantic structure (real text), $H_\ell < H(\mathbf{e}_t)$ and the gap $H(\mathbf{e}_t) - H_\ell$ grows with ℓ as long as topical context persists.

The corresponding working hypothesis is:

$$H_\ell^{\text{orig}} < H_\ell^{\text{FGN}} \approx H_\ell^{\text{shuf}} \approx H(\mathbf{e}_t), \quad \ell = 1, 2, \dots \quad (20)$$

The word-shuffled null is exchangeable and approaches the i.i.d. reference apart from finite-size dependence associated with fixed token counts; accordingly, $H_\ell^{\text{shuf}} \simeq H(\mathbf{e}_t)$. For the FGN surrogate, past ranks $r_{t-1}, \dots, r_{t-\ell}$ carry information about the present rank r_t (by the long-range memory of the FGN process); but the mapping $r \mapsto \phi(w_r)$ is semantically diffuse: words of similar Zipf rank span the full range of semantic content and are spread across all regions of the embedding space. Therefore past ranks carry negligible information about the present embedding $\mathbf{e}_t$, and $H_\ell^{\text{FGN}} \approx H(\mathbf{e}_t)$. For the original text, past embeddings directly encode the topical context of the narrative, providing genuine predictive information about subsequent embeddings.

6.2 A proxy measure: autocorrelation of projections

Direct estimation of H_ℓ in $\mathbb{R}^{300}$ requires large samples and density estimation in high dimensions, which is impractical for individual texts of length $M \sim 10^4$ – 10^5 . We therefore use a computationally tractable proxy: the normalised autocorrelation function (ACF) of the projected scalar series,

$$\rho_k(\ell) = \frac{\text{Cov}\left(x_t^{(k)}, x_{t+\ell}^{(k)}\right)}{\text{Var}\left(x_t^{(k)}\right)}, \quad (21)$$

for PCA component k and lag ℓ . For an i.i.d. process (word-shuffled null), $\rho_k(\ell) = 0$ for all $\ell \geq 1$ by construction. For the original text, $\rho_k(\ell) > 0$ with slow decay, reflecting the long-range topical structure. For the FGN surrogate, the entropy argument of Section 6.1 predicts $\rho_k(\ell) \approx 0$ despite the long-range correlations in the rank sequence.

Remark 4. *The ACF proxy is connected to the burstiness spectrum via the Wiener–Khinchine theorem: a slowly decaying $\rho_k(\ell)$ corresponds to excess power at low frequencies in the spectral density of $x_t^{(k)}$, which drives the temporal clustering of threshold crossings and hence large B_k . The burstiness spectrum and the ACF decay are therefore two complementary views of the same underlying temporal structure: the former is a non-linear (binary) summary, the latter a second-order (quadratic) characterisation.*

6.3 Empirical results: ACF of projections

Figure 8 shows $\rho_k(\ell)$ as a function of lag ℓ (log scale) for PC1, PC2, and PC3 of *Great Expectations* by Dickens ($\alpha = 0.689$), for the original text, the FGN surrogate, and the word-shuffled null. Figure 9 shows the same for the mean ACF over all $K = 20$ components.

Original text. The ACF is positive and significantly above the i.i.d. band $\pm 1.96/\sqrt{M} \approx \pm 0.008$ at lag $\ell = 1$ for all 20 components, with $\rho_k(1) \in [0.049, 0.208]$ (Table 7). The decay is gradual on a log scale, with values remaining above the i.i.d. band up to lags of order $\ell \sim 100$ for the lower-indexed components. This slow decay is a second-order signature of the temporal persistence that accompanies semantic burstiness.

FGN surrogate. All 20 components have $|\rho_k^{\text{FGN}}(1)| \leq 0.007$, inside the i.i.d. band. At lag $\ell = 100$ all values are inside the band. The FGN surrogate, despite matching the DFA exponent of the original ($\alpha = 0.689$), exhibits only negligible autocorrelation in the projected embedding series. This is the computational confirmation of equation (20): frequency-domain long-range memory does not propagate to semantic projections.

Word-shuffled null. As expected, $\rho_k^{\text{surr}}(\ell) \approx 0$ for all k and ℓ , consistent with the random-permutation reference.

Table 7: ACF of projections at lags $\ell = 1$ and $\ell = 100$ for *Great Expectations*, using the i.i.d. reference band ± 0.008 employed in the ACF calculation. Values outside the band are in bold.

k	$\rho_k(1)$		$\rho_k(100)$	
	Orig	FGN	Orig	FGN
1	0.060	0.004	0.008	0.004
2	0.208	0.007	0.015	0.004
3	0.136	-0.002	0.018	0.004
4	0.123	-0.003	0.008	-0.003
5	0.086	-0.002	0.004	-0.004
6	0.121	0.004	0.005	-0.001
7	0.127	0.002	0.007	-0.003
8	0.103	0.001	0.004	0.003
9	0.082	-0.002	0.008	0.001
10	0.083	0.004	0.003	0.003
$(k=11..20 \text{ all have } \rho_k^{\text{FGN}}(1) \leq 0.007)$				

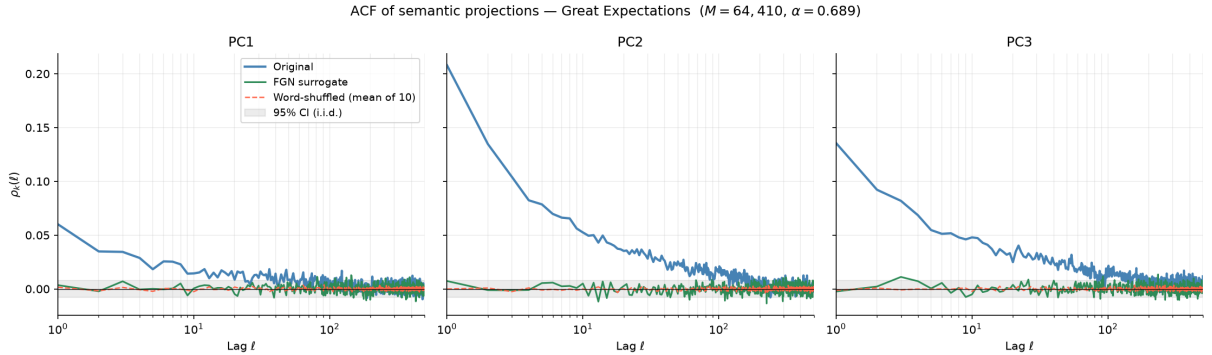


Figure 8: **Autocorrelation function $\rho_k(\ell)$ of projected series, as a function of lag ℓ (log scale) — *Great Expectations* (Dickens).** Each panel shows one PCA component (PC1, PC2, PC3). Blue: original filtered text. Green: FGN surrogate. Red dashed: word-shuffled null (mean over 10 realisations). Grey band: $\pm 1.96/\sqrt{M}$ i.i.d. reference (95% CI). The original text has substantial positive ACF that decays slowly on a log scale, remaining above the i.i.d. band up to $\ell \sim 100$. Both null models collapse immediately into the i.i.d. band, despite the FGN surrogate having the same DFA exponent $\alpha = 0.689$ as the original.

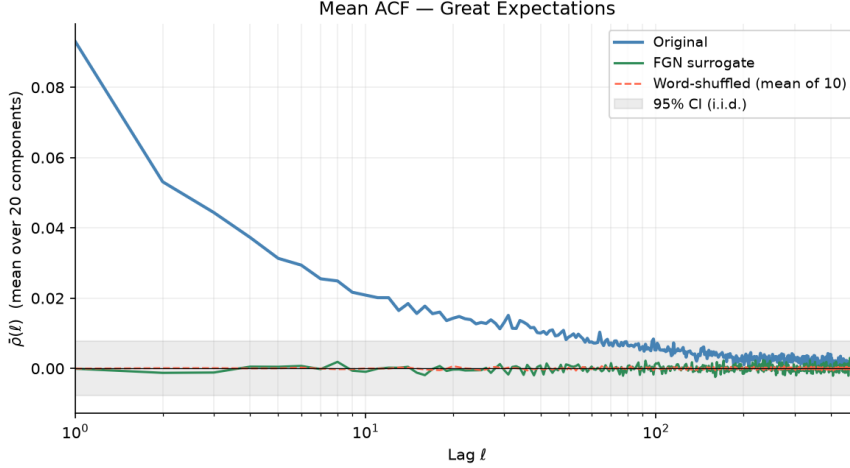


Figure 9: **Mean ACF $\bar{\rho}(\ell)$ averaged over all $K = 20$ PCA components — *Great Expectations* (Dickens).** Same colour conventions as Figure 8. The original text maintains positive mean autocorrelation well beyond the i.i.d. band across the full range of lags shown. The FGN surrogate and word-shuffled null are indistinguishable from the i.i.d. reference at all lags, confirming that frequency-domain long-range memory (present in the FGN) does not propagate to the semantic projection space.

6.4 Corpus-level ACF results

We computed $\rho_k(\ell)$ for all 16 texts and all $K = 20$ PCA components at lags $\ell \in \{1, 2, 5, 10, 20, 50, 100, 200, 300\}$, yielding $16 \times 20 \times 9 = 2,880$ measurements. The 95% confidence band for an i.i.d. process at sample size M is $\pm 1.96/\sqrt{M}$; since M varies across texts, we use the text-specific CI for significance tests.

Mean ACF decay. Figure 10 shows the mean $\rho(\ell)$ averaged over all (text, k) pairs as a function of lag. The original text maintains positive and significant autocorrelation across the full range of lags tested, with mean values

$$\bar{\rho}^{\text{orig}}(1) = 0.100 \pm 0.045, \quad \bar{\rho}^{\text{orig}}(10) = 0.032 \pm 0.021, \quad \bar{\rho}^{\text{orig}}(100) = 0.011 \pm 0.011, \quad (22)$$

where $\pm$ denotes one standard deviation over (text, k) pairs. The FGN surrogate and the word-shuffled null are indistinguishable from each other and from zero across all lags:

$$\bar{\rho}^{\text{FGN}}(1) = 0.002 \pm 0.005, \quad \bar{\rho}^{\text{sur}}(1) = 0.000 \pm 0.004. \quad (23)$$

The ratio $\bar{\rho}^{\text{orig}}(1)/\bar{\rho}^{\text{FGN}}(1) \approx 50$ quantifies how completely the FGN rank-to-embedding mapping destroys the frequency-domain long-range correlations in semantic projection space.

Fraction of significant pairs. Figure 11 shows, for each lag, the fraction of (text, k) pairs where $|\rho_k(\ell)|$ exceeds the i.i.d. band. At lag $\ell = 1$: 100% of original-text pairs are significant, 10.9% of FGN pairs, and 4.7% of shuffled pairs. The FGN figure exceeds the expected false-positive rate of 5%, but the significant cases are concentrated on PC1 (56% of the PC1 pairs, against 8% for $k \geq 4$) — the component most affected by the geometric artefact of rare transliterated proper nouns in the GloVe embedding space (Section 3.1). From lag $\ell = 2$ onwards the FGN fraction lies between 2.5% and 7%, at the false-positive

level. The effect is in any case tiny relative to the separation of the original projections from both controls.

The “original only” curve (orig significant, FGN not) peaks at 93% at lag $\ell = 2$ and remains above 50% up to lag $\ell = 100$, providing a clean corpus-level version of the single-text result. The original text maintains detectable semantic autocorrelation at scales of order 100 tokens, corresponding to passages of roughly a paragraph, consistent with the topical structure of literary narrative.

Distribution of $\rho_k(1)$. Figure 12 shows the distribution of $\rho_k(1)$ over all 320 (text, k) pairs. Panel A shows two completely separated distributions: the original has $\rho_k(1) \in [0.026, 0.264]$ with mean 0.100, all pairs significant; the FGN has $\rho_k(1) \in [-0.014, 0.023]$ with mean 0.002, the vast majority inside the i.i.d. band.

Panel B zooms on the FGN and shuffled distributions. Both are centred near zero and overlap substantially. A one-sample t -test of $\bar{\rho}^{\text{FGN}}(1) = 0$ gives $t = 5.05$, $p < 10^{-6}$: the mean is statistically non-zero but negligibly small (0.002) relative to the orig–null gap (0.098). As with the burstiness differences (Section 5.4), the statistical significance at this sample size reflects estimator sensitivity rather than a scientifically meaningful effect.

Remark 5. *The PC1 anomaly — where $\rho_1^{\text{FGN}}(1)$ is occasionally significant — is a known limitation of the current pipeline. PC1 is partially dominated by a geometric cluster of low-frequency words in GloVe that are distant from the main vocabulary cloud (proper nouns, transliterations, archaic forms). Both the original text and the FGN surrogate generate sequences that alternate between this cluster and the main cloud, producing spurious short-range autocorrelation in PC1 for both. This artefact is independent of topical burstiness and disappears when the frequency filter is tightened to $R = 150$ or higher. All results for $k \geq 2$ are unaffected.*

6.5 DFA of projected series

The ACF results of Sections 6.3 and 6.4 confirm the predicted ordering $\rho_k^{\text{orig}}(\ell) \gg \rho_k^{\text{FGN}}(\ell) \approx \rho_k^{\text{surr}}(\ell) \approx 0$. We now complement the ACF analysis by computing the DFA scaling exponent of the projected series $x_t^{(k)}$ directly, providing the semantic-level analogue of the symbolic-level DFA result that motivates the paper.

For each PCA component k and each of the three series (original, FGN, word-shuffled), we apply DFA to the scalar time series $x_t^{(k)}$ and estimate the scaling exponent α_k from $F(s) \sim s^{\alpha_k}$, where $F(s)$ is the detrended fluctuation function at box size s (Appendix A). Box sizes range from $s_{\min} = 10$ to $s_{\max} = M/4$ on a logarithmic grid.

Results on Great Expectations. Table 8 and Figure 13 report α_k for all $K = 20$ components of *Great Expectations* by Dickens ($\alpha = 0.689$). The results confirm the prediction precisely:

- **Original:** $\alpha_k^{\text{orig}} > 0.5$ in all 20/20 components, with mean $\bar{\alpha}^{\text{orig}} = 0.623 \pm 0.034$. All values exceed the shuffled reference by more than 2σ , confirming long-range correlations in every semantic projection.
- **FGN surrogate:** $|\alpha_k^{\text{FGN}} - 0.5| < 0.05$ in all 20/20 components, with mean $\bar{\alpha}^{\text{FGN}} = 0.499 \pm 0.014$. Despite having a matched DFA exponent ($\alpha = 0.689$, generator

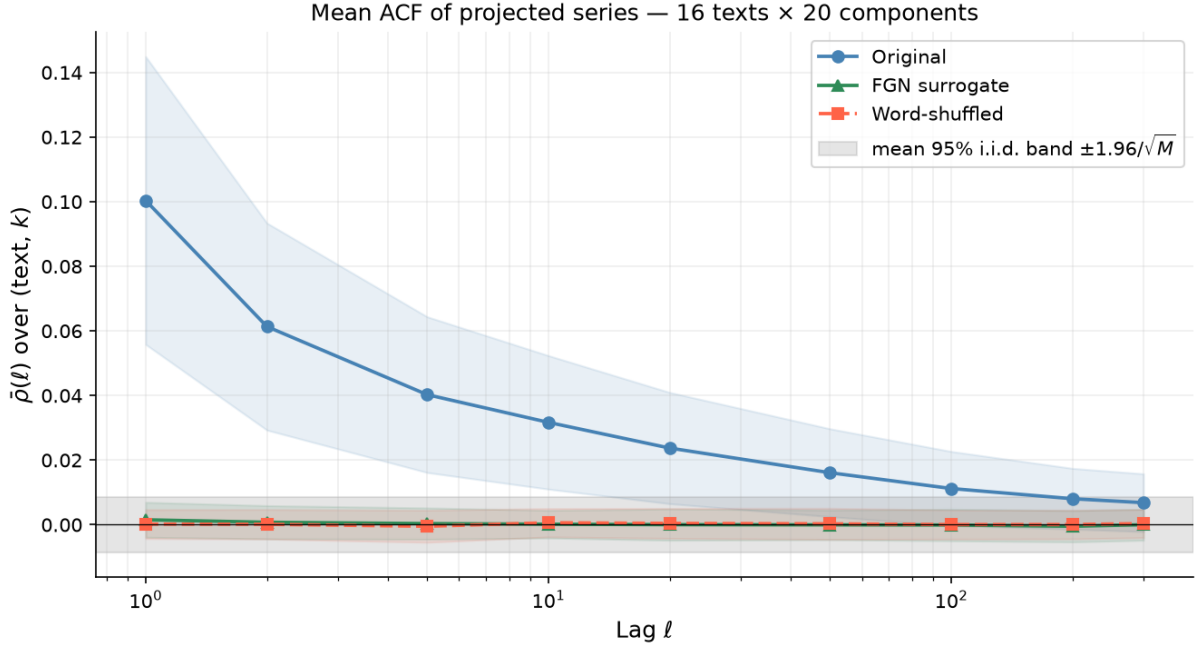


Figure 10: **Mean ACF $\bar{\rho}(\ell)$ of projected series across the corpus of 16 texts and $K = 20$ components, as a function of lag ℓ (log scale).** Shaded bands: ± 1 standard deviation over (text, k) pairs. Grey band: mean 95% i.i.d. confidence band $\pm 1.96/\sqrt{M}$. Blue circles: original filtered text. Green triangles: FGN surrogate. Red squares: word-shuffled null. The original maintains significant positive ACF across all lags tested ($\ell = 1, \dots, 300$). Both null models are indistinguishable from zero at all lags.

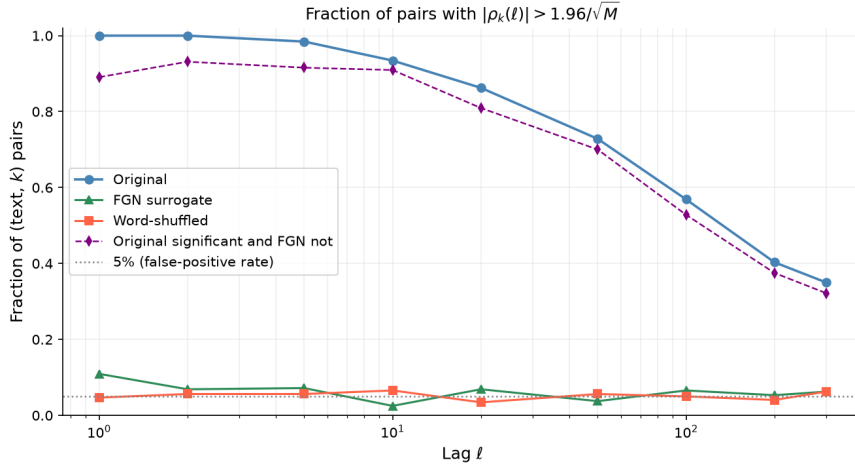


Figure 11: **Fraction of text–component pairs with significant ACF as a function of lag ℓ (log scale).** Blue: original; green: FGN; red: word-shuffled; purple dashed: original significant and FGN not. The original remains clearly separated from both controls through lags of order 10^2 . The short-lag FGN excess is concentrated disproportionately in PC1, the component most affected by the embedding-geometry artefact discussed in Section 3.1.

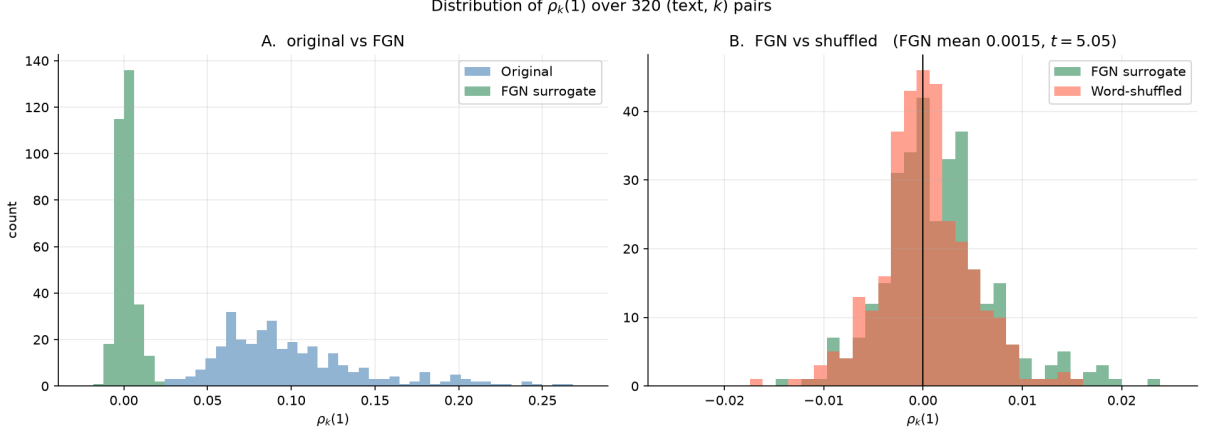


Figure 12: **Distribution of $\rho_k(1)$ over 320 (text, k) pairs.** **Panel A:** original (blue) and FGN (green). The two distributions are completely separated: all original pairs have $\rho_k(1) > 0.007$ (above the i.i.d. band), while nearly all FGN pairs are inside the band. **Panel B:** FGN (green) and word-shuffled (red) on an expanded scale. Both are centred near zero; a t -test gives $t = 5.05$ for $\bar{\rho}^{\text{FGN}}(1) = 0$ but the effect size is negligible (0.002) relative to the orig–null gap (0.098).

exponent $\alpha_0 = 0.74$) in the rank sequence by construction, the FGN projects onto every semantic direction as an uncorrelated process.

- **Word-shuffled:** $\bar{\alpha}^{\text{surr}} = 0.499 \pm 0.004$, consistent with $\alpha = 0.5$ (white noise) as expected for an i.i.d. sequence.

Table 8: DFA scaling exponents α_k of projected series $x_t^{(k)}$ for *Great Expectations* (Dickens, $\alpha = 0.689$). Asterisk (*): α_k^{orig} exceeds the shuffled reference by $> 2\sigma$. Tilde (~): $|\alpha_k^{\text{FGN}} - 0.5| < 0.05$.

k	α^{orig}	α^{FGN}	α^{surr}	k	α^{orig}	α^{FGN}	α^{surr}
1	0.605*	0.511~	0.501	11	0.575*	0.506~	0.500
2	0.681*	0.500~	0.491	12	0.642*	0.493~	0.502
3	0.693*	0.518~	0.495	13	0.564*	0.512~	0.500
4	0.634*	0.514~	0.499	14	0.595*	0.484~	0.497
5	0.642*	0.486~	0.496	15	0.645*	0.486~	0.497
6	0.633*	0.515~	0.501	16	0.579*	0.505~	0.494
7	0.647*	0.504~	0.498	17	0.586*	0.511~	0.493
8	0.633*	0.469~	0.497	18	0.617*	0.515~	0.507
9	0.643*	0.482~	0.494	19	0.595*	0.489~	0.505
10	0.630*	0.482~	0.503	20	0.622*	0.495~	0.502
Mean			$\bar{\alpha}^{\text{orig}} = 0.623 \pm 0.034$	$\bar{\alpha}^{\text{FGN}} = 0.499 \pm 0.014$			$\bar{\alpha}^{\text{surr}} = 0.499 \pm 0.004$

Figure 14 shows the log-log fluctuation function $F(s)$ for PC1, PC2, and PC3. The original text shows a clear power-law scaling with slope $\alpha_k \approx 0.60$ – 0.69 , visibly steeper than the reference slope 0.5. The FGN and shuffled curves are parallel to the reference slope, confirming $\alpha_k \approx 0.5$ for both null models.

Remark 6. The exponents $\alpha_k^{\text{orig}} \in [0.564, 0.693]$ do not decrease monotonically with k . Some higher-indexed components (e.g. PC12: $\alpha = 0.642$, PC15: $\alpha = 0.645$) have stronger long-range correlations than some lower-indexed ones (e.g. PC11: $\alpha = 0.575$, PC13: $\alpha = 0.564$). This partial decorrelation between PCA rank (ordered by variance) and DFA

exponent mirrors the decorrelation between the eigenvalue spectrum $\{\lambda_k\}$ and the burstiness spectrum $\{B_k\}$ reported in Section 4: semantic variance and semantic temporal structure are distinct properties of narrative organisation.

Consistent temporal signatures. The DFA, ACF, and burstiness analyses provide mutually consistent but non-equivalent summaries of temporal organisation. For the original projections, $\alpha_k > 0.5$, the ACF remains positive over extended lags, and B_k exceeds the geometric reference. For both controls, $\alpha_k \simeq 0.5$, the ACF is close to zero, and $B_k \simeq CV_{\text{geom}}$. These observations can be summarised as

$$\alpha_k^{\text{orig}} > 0.5, \quad \rho_k^{\text{orig}}(\ell) > 0 \text{ over extended lags}, \quad B_k^{\text{orig}} > CV_{\text{geom}}, \quad (24)$$

whereas the FGN and shuffled projections remain close to their uncorrelated references in all three measures.

The FGN surrogate is especially informative because its rank sequence has the prescribed long-range DFA exponent, while every analysed semantic projection has $\alpha_k^{\text{FGN}} \simeq 0.5$. Rank-level persistence therefore does not propagate automatically into semantic projection space.

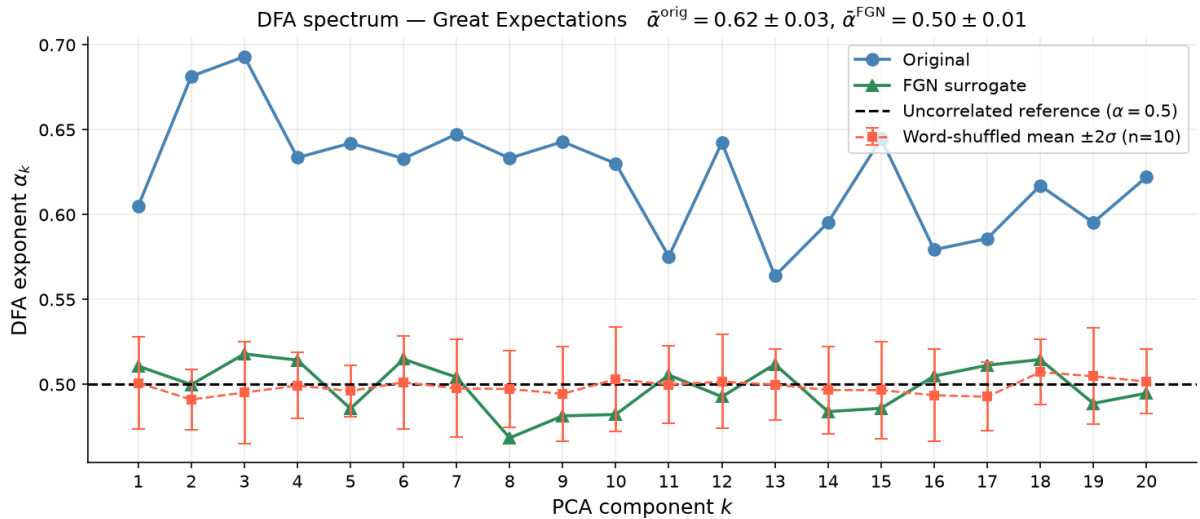


Figure 13: DFA scaling exponent α_k of projected series $x_t^{(k)}$ as a function of PCA component k — *Great Expectations* (Dickens). Blue circles: original filtered text. Green triangles: FGN surrogate. Red squares: word-shuffled null (mean $\pm 2\sigma$ over 10 realisations). Dashed line: uncorrelated reference $\alpha = 0.5$. The original text has $\alpha_k > 0.5$ for all 20 components; the FGN and shuffled are indistinguishable from $\alpha = 0.5$.

6.6 Connection between ACF and burstiness

The ACF and the burstiness parameter probe different aspects of the same projected series. By the Wiener–Khinchine theorem, slow ACF decay is associated with excess low-frequency power in the spectral density

$$S_k(\omega) = \sum_{\ell=-\infty}^{\infty} \rho_k(\ell) e^{-i\omega\ell}. \quad (25)$$

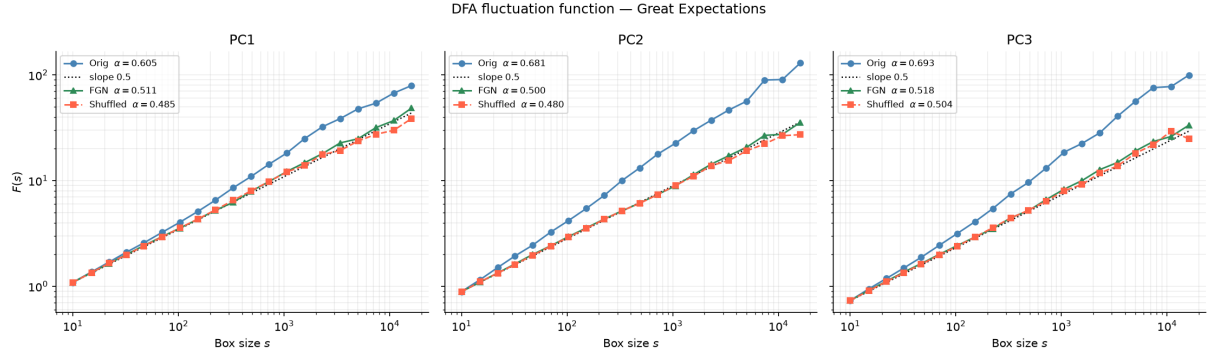


Figure 14: **Log-log DFA fluctuation function $F(s)$ for PC1, PC2, and PC3 — *Great Expectations* (Dickens).** Each panel shows one component; the estimated exponent α_k is given in the legend. Dotted line: reference slope 0.5. The original text (blue) has a consistently steeper slope than both null models, which are parallel to the $\alpha = 0.5$ reference across all box sizes s .

Such low-frequency variation is consistent with extended semantic episodes and clustered threshold exceedances, and therefore with broad IET distributions. The connection is not one-to-one, however: thresholding is non-linear, and B_k depends on temporal structure beyond the second-order information contained in the ACF.

The empirical conclusion is consequently based on the agreement of three complementary observations rather than on a deterministic causal identity. Original texts show elevated B_k , positive projection ACFs, and projection DFA exponents above 0.5. The FGN and word-shuffled controls remain close to the geometric, zero-ACF, and white-noise references, respectively. This convergence of measures supports the interpretation that ordered topical organisation, rather than global rank-level scaling alone, produces persistence in semantic projection space.

7 Discussion and Conclusions

CRedit authorship contribution statement

Mirko Degli Esposti: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Marcelo A. Montemurro:** Conceptualization, Methodology, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work the authors used a large language model to assist with code refactoring and manuscript editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

All texts are public-domain works from Project Gutenberg. The corpus, the analysis code, the result tables and the scripts that regenerate every figure in this paper are available at <https://github.com/mirko-degli-esposti/semantic-burstiness-texts>. Pre-trained GloVe vectors are distributed by Stanford NLP (Pennington et al., 2014).

Acknowledgements

A Geometric distribution as IET null: derivation

Consider a binary sequence $b_1, b_2, \dots, b_M$ where each b_t is independently Bernoulli(p). An *event* occurs at position t when $b_t = 1$. The inter-event time τ between two successive events is the number of positions (including the terminal event) from one event to the next. The probability of $\tau = k$ requires $b_{t+1} = \dots = b_{t+k-1} = 0$ and $b_{t+k} = 1$:

$$\Pr(\tau = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots \quad (26)$$

which is the geometric distribution with parameter p . Its mean, variance, and coefficient of variation are

$$\langle \tau \rangle = \frac{1}{p}, \quad \text{Var}(\tau) = \frac{1 - p}{p^2}, \quad \text{CV}_{\text{geom}}(p) = \sqrt{1 - p}. \quad (27)$$

The exponential distribution $p(\tau) \propto e^{-\lambda\tau}$ arises in the limit $p \rightarrow 0$ with $\lambda = -\log(1-p) \approx p$, and has $\text{CV} = 1$ (Poisson process). For $p = 0.05$, $\text{CV}_{\text{geom}} = 0.975 \approx 1$, so the Poisson approximation is adequate. For $p = 0.5$, $\text{CV}_{\text{geom}} = 0.707$, and the Poisson approximation fails badly.

B Additional per-text figures

Eigenvalue spectra and inter-event-time distributions for the three representative texts other than *War and Peace*, whose corresponding figures appear in the main text (Figures 1 and 2). Equivalent figures for every text of the corpus, at any threshold quantile, can be regenerated with `scripts/run_text.py` in the code repository.

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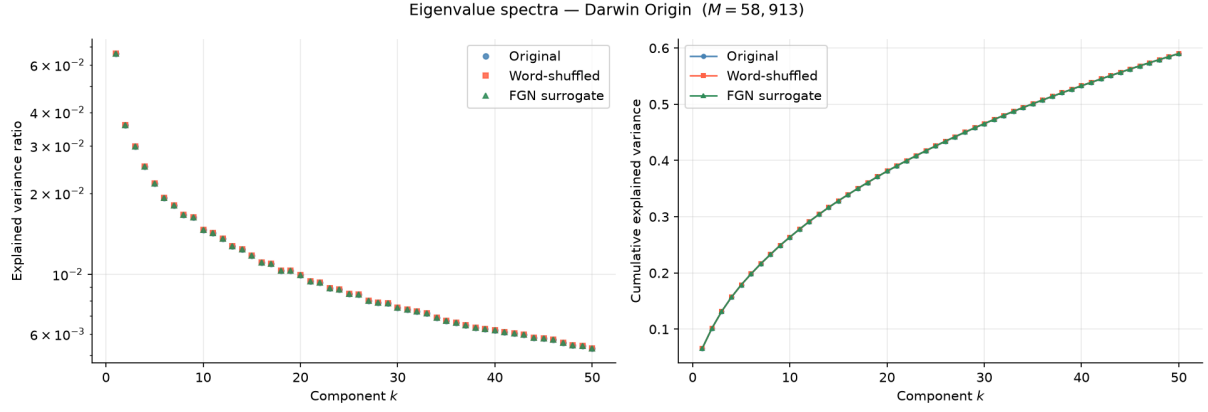


Figure 15: Eigenvalue spectra and cumulative explained variance for *On the Origin of Species*. Same conventions as Figure 1.

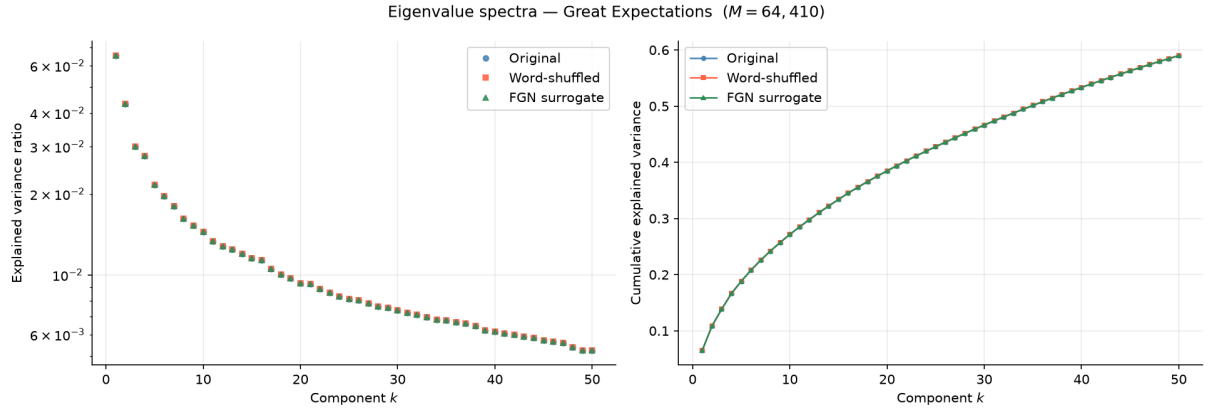


Figure 16: Eigenvalue spectra and cumulative explained variance for *Great Expectations*. Same conventions as Figure 1.

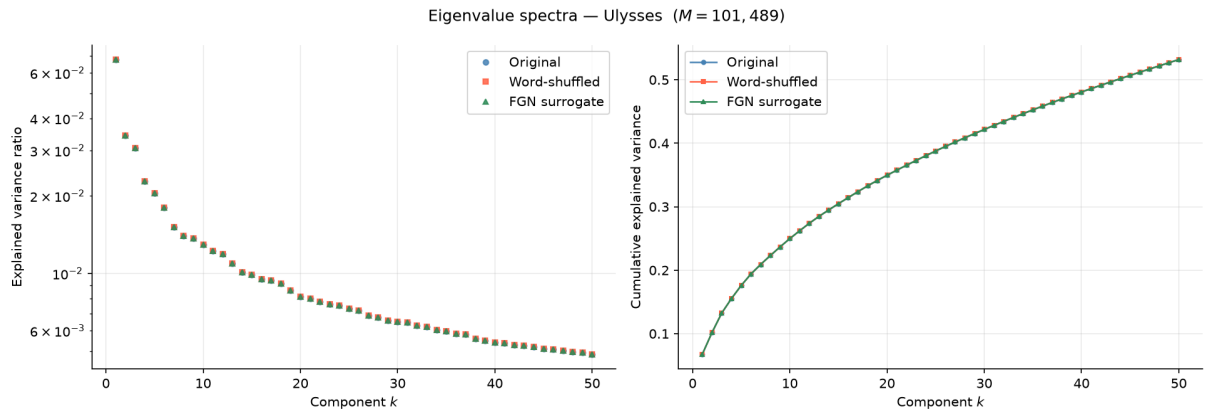


Figure 17: Eigenvalue spectra and cumulative explained variance for *Ulysses*. The more gradual eigenvalue decay reflects the greater stylistic diversity of the 18 episodes compared to the other texts.

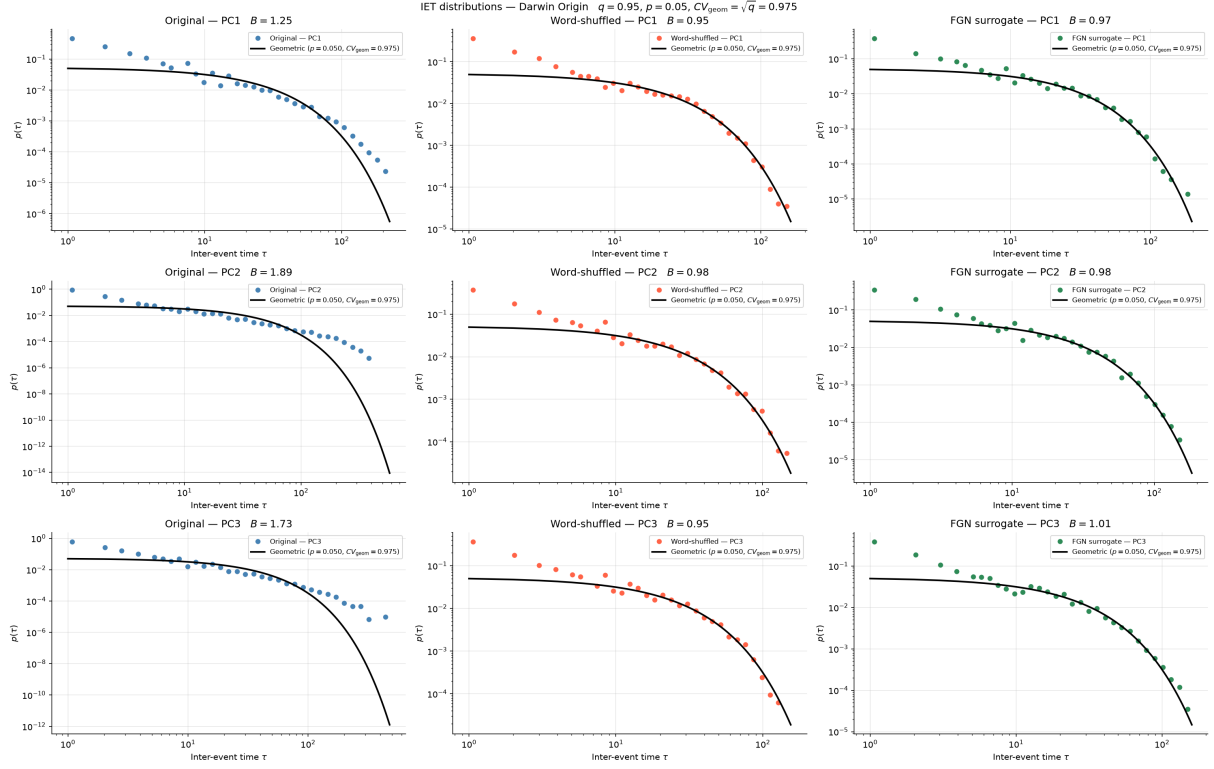


Figure 18: Inter-event time distributions for PC1, PC2, PC3 of *On the Origin of Species* ($q = 0.95$). Same conventions as Figure 2.

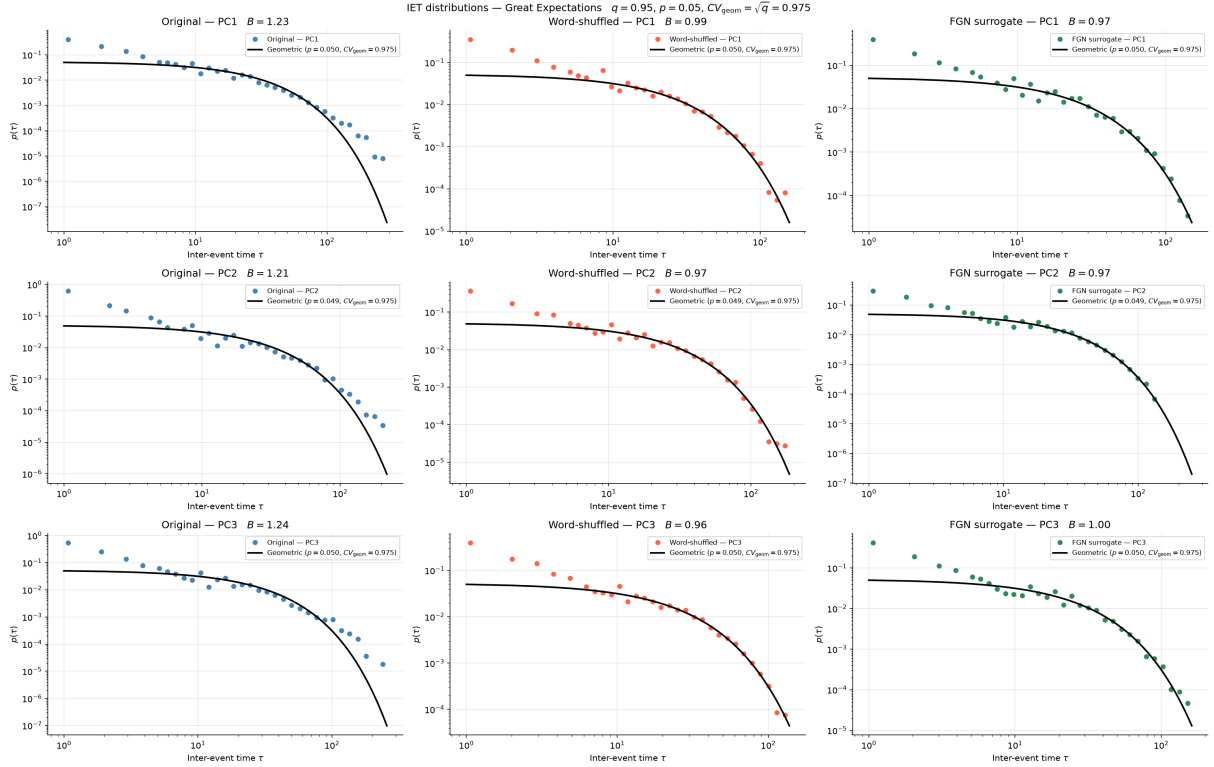


Figure 19: Inter-event time distributions for PC1, PC2, PC3 of *Great Expectations* ($q = 0.95$). Same conventions as Figure 2.

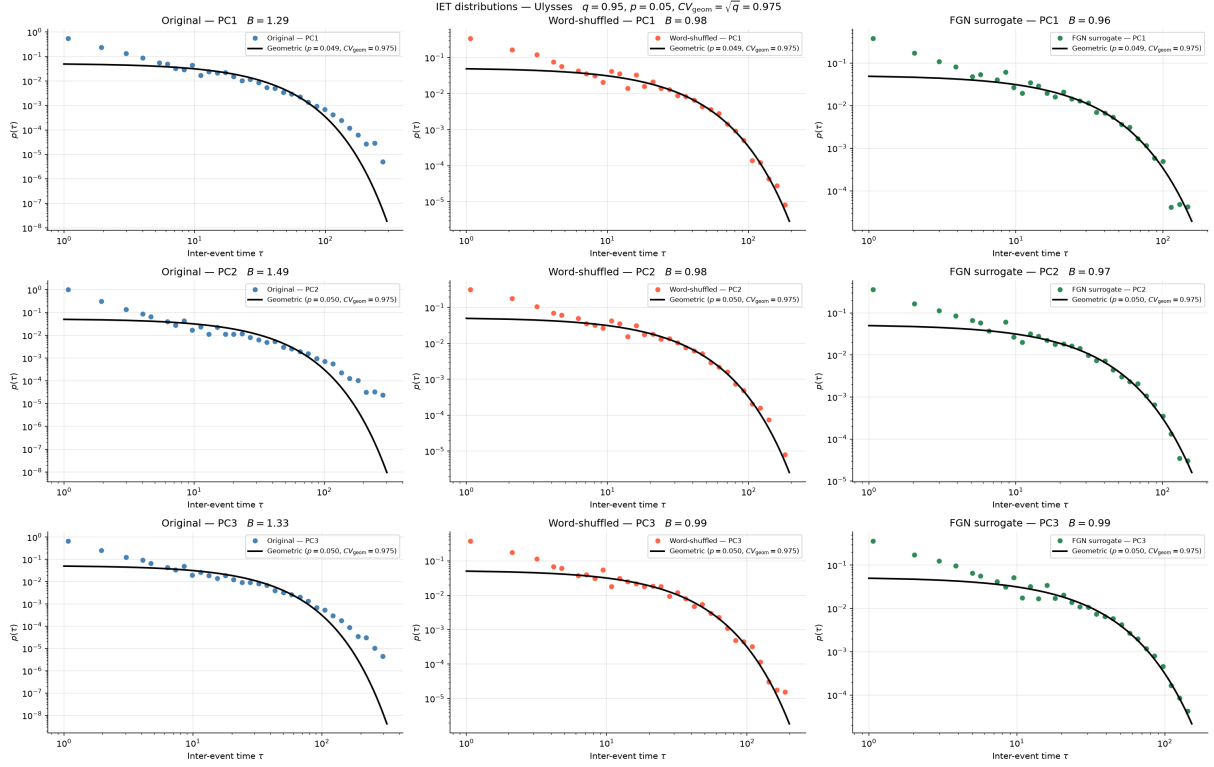


Figure 20: Inter-event time distributions for PC1, PC2, PC3 of *Ulysses* ($q = 0.95$). Same conventions as Figure 2.

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